

Chapter 9. Pressure-Controlled and Inverse-Ratio Ventilation

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Pressure-Controlled and Inverse-Ratio Ventilation: Introduction

The use of pressure-controlled ventilation (PCV) increased substantially after 1995, when intensivists became increasingly aware of ventilator-induced lung injury (VILI) and the risks of high inspiratory pressures. Familiarity with the concept of permissive hypercapnia contributed to this change, helping physicians to overcome the old and stringent mindset on arterial blood gases.^{1–4} Tidal volume or minute ventilation requirements were increasingly regarded as secondary goals during mechanical ventilation and the apparent security provided by PCV, keeping airway pressures under strict limits, gained broader acceptance. Recent surveys in intensive care units demonstrate that PCV is now used in up to 25% of ventilated patients, usually in the most severe cases^{5,6} (including pediatric patients⁷). Studies, describing the implications of PCV on the cardiovascular system, work of breathing, regional mechanics, risks of VILI, and recruitment maneuvers are now available. These studies increase physician comfort in moving away from volume-controlled ventilation.

In contrast to the increasing acceptance of PCV, the use of inverse-ratio ventilation (IRV)—that is, the prolongation of inspiratory time to the point of inverting the conventional inspiratory-to-expiratory (I:E) ratio—is now very rare. This decline was caused by recent progress in our understanding on the pathophysiology of lung collapse, and demonstration of unnecessary hemodynamic compromise. The original benefits in terms of oxygenation ascribed to IRV,^{8–14} and the apparent reduction of peak inspiratory alveolar pressures^{11,12} are now supplanted by more consistent effects of recruiting maneuvers followed by optimization of positive end-expiratory pressure (PEEP),^{15–20} especially during controlled mechanical ventilation. It is only in the more specific context of airway pressure release ventilation (APRV) that the use of IRV has still survived, although surrounded by great skepticism and controversy.^{21,22}

In this chapter, we do not revitalize IRV. We believe there are always safer, more predictable, and more efficient ventilatory solutions to be implemented at the bedside. IRV was extensively discussed in the second edition of this book. At the end of the present chapter, we present a few aspects of IRV that still offer great insights about the pathophysiology of acute respiratory failure.

Types of Controlled Ventilation and Selection of the “Controlled” Parameter

Ventilators regulate the pressure profile applied to the airways or the pattern of flow delivery. Somewhat imprecisely, flow-controlled ventilation has been designated “volume-controlled” ventilation (VCV). We avoid this convention because the criterion by which the ventilator ceases to pressurize the airway (initiates deflation) may be a specified value of delivered volume, pressure, elapsed time, or flow. The variable, however, actively controlled by the ventilator during “volume-controlled” breaths is in reality inspiratory flow. Therefore, the modes of ventilation currently used in medical practice should be classified as *pressure-controlled* or *flow-controlled* and as *time-cycled*, *volume-cycled*, *flow-cycled*, or *pressure-cycled*. Pressure-support ventilation (PSV) is an example of a pressure-controlled, *flow-cycled* mode, whereas PCV is an example of pressure-controlled, *time-cycled* mode.

In flow-controlled modes, the waveform theoretically can be of any desired contour; in practice, however, the flow waveforms usually are rectilinear (square), linearly decelerating, and sinusoidal. Setting tidal volume as the “off switch” criterion (*volume-cycled*) means that the pressure applied to the airway opening can rise to any value required by impedance to inflation that does not exceed the pressure-limit alarm. For instance, very high absolute airway pressures can develop during acute airway obstruction, acute lung edema or expulsive efforts or bouts of coughing, without consequences to delivered tidal-volume (except if the alarm is activated).

Although once used extensively, *pressure-cycled* ventilation is now considered obsolete for continuous support, and its application in intermittent positive-pressure breathing has been restricted greatly as a backup cycling-off criterion during flow-controlled, volume-cycled ventilation (when airway pressures reach a preset alarm threshold), or as a backup for flow-cycling during pressure support ventilation.

To improve safety or decrease the need for repeated adjustments at the bedside, flow-controlled and pressure-controlled algorithms have been combined recently to form new modes (*dual modes*, such as “volume-assured pressure support” and “pressure-regulated volume control”), incorporating the desirable characteristics of each category.^{23,24}

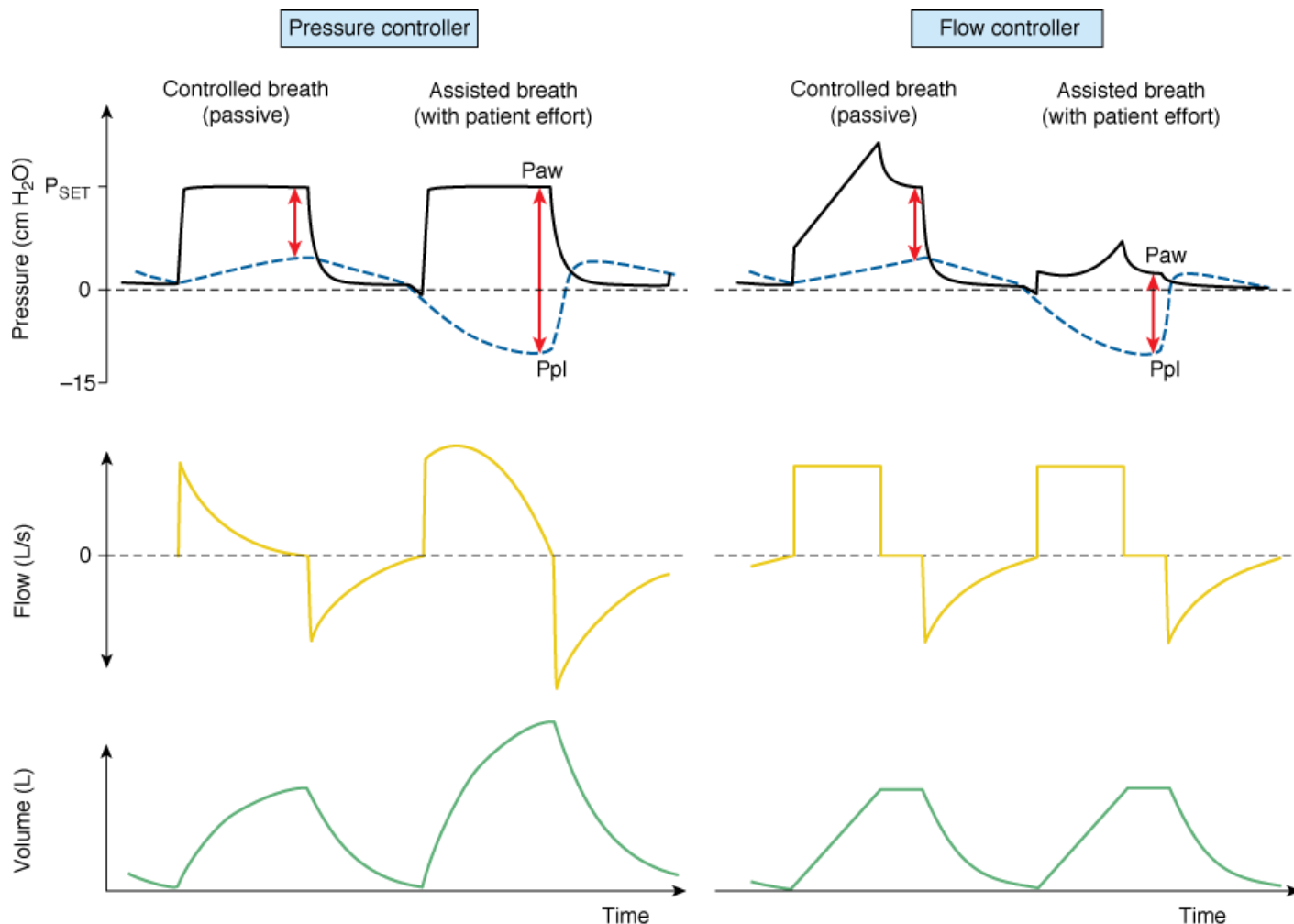
The equation of motion of the respiratory system confines the clinician to setting independently the inspiratory flow and tidal volume or just the applied pressure profile. Flow and pressure cannot be selected independently at the same time because their relationship is intrinsically defined by the mechanical properties of the respiratory system. Because modern ventilators can provide online feedback information about output variables (airway pressures during flow-controlled ventilation or flow and tidal volume during pressure-controlled ventilation), either flow-controlled, *volume-cycled* ventilation or pressure-controlled, *time-cycled* ventilation (**PCV**) can be used interchangeably during specific conditions. For instance, instead of using the original pressure-control algorithms during IRV, some investigators proposed the equivalent use of IRV through flow-controlled, *volume-cycled* breaths,¹³ although doing so requires careful monitoring and frequent readjustments. Depending on particular combinations of compliance and resistance in the respiratory system, the pressure profile generated in the airways can be similar in both modes.

The major differences between *flow-controlled* and *pressure-controlled* breaths appear during prolonged use, after unpredictable changes in respiratory system characteristics. By using the traditional flow-controlled, *volume-cycled* mode, minute ventilation is guaranteed safely over prolonged periods of time, although airway pressures may rise significantly when respiratory system impedance increases. Conversely, during prolonged use of pressure-controlled, *time-cycled* ventilation, airway pressure limits are guaranteed, although minute ventilation is at risk whenever lung impedance changes.

Independent of selection of the controlled parameter, inspiration can be triggered by elapsed time or by small perturbations in pressure and flow in the airways (usually indicative of patient effort). Elapsed time defines a totally controlled breath, whereas small perturbations in pressure and flow define an assisted (patient-triggered) breath. In recent years, modern ventilators have incorporated algorithms providing pressure-controlled breaths triggered by either elapsed expiratory time, characterizing traditional **PCV**, or patient effort (*assisted* pressure-controlled ventilation), analogous to the traditional flow-controlled assist-control ventilation.

During *assisted PCV*, some important and relevant differences between VCV and **PCV** may appear. For instance, in the presence of a sustained inspiratory effort, extending from the beginning until end-inspiration, the *pressure*-controller will respond with an increased delivery of flow and tidal volume so as to keep airway pressures close to target. The result is an increased transpulmonary effective pressure (which is the sum of the positive airway pressure generated by the ventilator plus the modulus of the negative perturbation in pleural pressure driven by inspiratory muscles). In contrast, the *flow*-controller keeps inspiratory flow close to the target, while peak airway pressures decrease in proportion to the negative perturbation in pleural pressures. The result is a near constant transpulmonary effective pressure (**Fig. 9-1**, which shows that the modulus of positive airway pressure from the ventilator decreases in direct proportion to the modulus of the negative perturbation in pleural pressures; thus, the sum of the two is almost constant). This is why some investigators believe that VCV is safer during acute lung injury in the presence of strong patient effort and a persisting need for lung protection.^{25,26} The drawback of this “forced” maintenance of transpulmonary pressures by VCV, however, is the increased workload to inspiratory muscles and the promotion of a regimen of more negative pleural pressures, which, when excessive, may promote lung edema.²⁷

Figure 9-1



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Airway opening pressure (P_{aw}), pleural pressure (P_{pl} —blue dashed lines), flow and volume tracings during pressure-controlled ventilation (PCV, left panel of curves) and during flow-controlled ventilation (VCV, right panel of curves). In this simulation, pleural pressures are +1 cm H₂O at relaxed end expiration, which corresponds to the midpleural pressure in a supine patient under anesthesia. In the second cycle of each panel, a patient effort is simulated (muscle pressure = -15 cm H₂O). Red arrows at end-inspiration represent the total amplitude of effective transpulmonary pressure at end-inspiration. Note that transpulmonary pressures may increase during assisted PCV, in conjunction with increases in inspiratory flow and tidal volume (second breath in the left panel). In contrast, the assisted-VCV breath (second breath in the right panel) is associated with a drop in airway pressures, and with sustained inspiratory flow and tidal volume. The result is a near-constant transpulmonary pressure at end-inspiration. Because of safety concerns, some modern ventilators do not operate as pure flow controllers during VCV if airway pressure drops to below baseline pressure (PEEP); they may provide an extra demand flow to compensate excessive drops in Paw. In this case, the effective transpulmonary pressure might also increase during VCV, although to a lesser degree.

Specific Characteristics of Pressure-Controlled Ventilation

Many recently introduced ventilator modes can be considered as variants of pressure-controlled or *pressure-preset* ventilation. This includes traditional PCV,^{8,27} PSV,^{28,29} biphasic continuous positive airway pressure, and variants of APRV.^{30,31} In some modes, spontaneous respiratory efforts continue. In others, none occur. These modes vary both in their intended objectives and in their criteria for initiating and terminating the machine's inspiratory cycle. Yet all can be viewed as modes in which the machine applies approximately square waves of pressure to the airway opening. In concept, any pressure profile can be regulated. In current practice, however, most *pressure-controlled* modes build pressure rapidly, toward a preset value, attempting to maintain pressure nearly constant throughout the remainder of the higher-pressure phase. During exhalation, pressure is released abruptly, allowing passive deflation to occur unimpeded or against a set PEEP level. By fixing the level of applied pressure during inspiration, the physician imposes an upper limit to the machine's energy transferred to lung tissue.

The specifics of the new pressure-controlled modes are discussed in other chapters. The physical principles governing pressure-controlled, *time-cycled* ventilation—which serves as the prototype for this group—and its major “outcome” variables, such as tidal volume, minute ventilation, and intrinsic PEEP, are addressed here. Some discussion of the concepts of mean airway pressure ($\bar{P}_{aw}$) and mean alveolar pressure ($\bar{P}_A$) also are presented because these concepts are essential tools for understanding the physiologic implications of all pressure-controlled modes.

Input Parameters of Pressure-Controlled Ventilation

Apart from the external PEEP, the clinician sets only three parameters during PCV: the applied inspiratory pressure, backup or mandatory frequency, and fractional inspiratory time (duty cycle, T_I/T_{TOT}). The most salient feature of PCV is that maximal airway and alveolar pressures are restricted by the cap of preset pressure, whereas tidal volume, flow, minute volume, and alveolar ventilation depend on the impedance of the respiratory system in conjunction with the three input variables just described. Once the impedance (resistance and compliance) of the respiratory system is known, the machine’s contributions to ventilation and alveolar pressure can be characterized completely from knowledge of just those three input parameters.

Machines vary in the rapidity (*rise time* or *inspiratory slope*) with which airway pressure builds toward the preset maximum value. Faster rates of pressurization are needed in certain situations when flow demands are high (during assisted *pressure-controlled* ventilation) so as to achieve high peak inspiratory flows that exceed patient demands. Because of limitations in the controlling system of most ventilators, however, a faster (shorter) *rise time* has to be selected during low-impedance conditions (in large patients), whereas a slower one has to be selected during high-impedance conditions (to avoid overshoots in airway pressure at the beginning of inspiration). Some newer machines allow the clinician to adjust rise time to suit the situation at hand, whereas others adjust it automatically.

With most machines today, physicians set the pressure increment above external PEEP to be applied during inspiration. This means that absolute inspiratory pressures increase as external PEEP increases. With a few machines, the physician has to set the absolute inspiratory pressure, which is independent of external PEEP.

Mean Airway and Alveolar Pressures

When considering a plot of airway-opening pressure (P_{aw}) over time, mean airway pressure is the integral of P_{aw} over time divided by the time span of the breath. Two input parameters that the clinician sets during PCV bear direct relationships to mean airway pressure ($\bar{P}_{aw}$): preset inspiratory pressure and T_I/T_{TOT} . Because of the square waveform of pressure during PCV, $\bar{P}_{aw}$ can be expressed simply as

$$\bar{P}_{aw} = P_{SET} \times T_I / T_{TOT} + PEEP_E \times (1 - T_I / T_{TOT}) \quad (1)$$

where $PEEP_E$ is external PEEP. Therefore, variations in either set pressure (P_{SET}) or T_I/T_{TOT} influence $\bar{P}_{aw}$ predictably. Frequency variations leave $\bar{P}_{aw}$ unaffected.^{28,29}

Under passive conditions, airway pressure represents the total pressure applied across the respiratory system. Under this circumstance, it can be demonstrated mathematically that mean alveolar pressure $\bar{P}_A$ bears a close relationship with $\bar{P}_{aw}$, and its bedside estimation requires only some previous assessment of inspiratory and expiratory resistances

$$\bar{P}_A = \bar{P}_{aw} + \dot{V}_E \times (R_E - R_I) \quad (2)$$

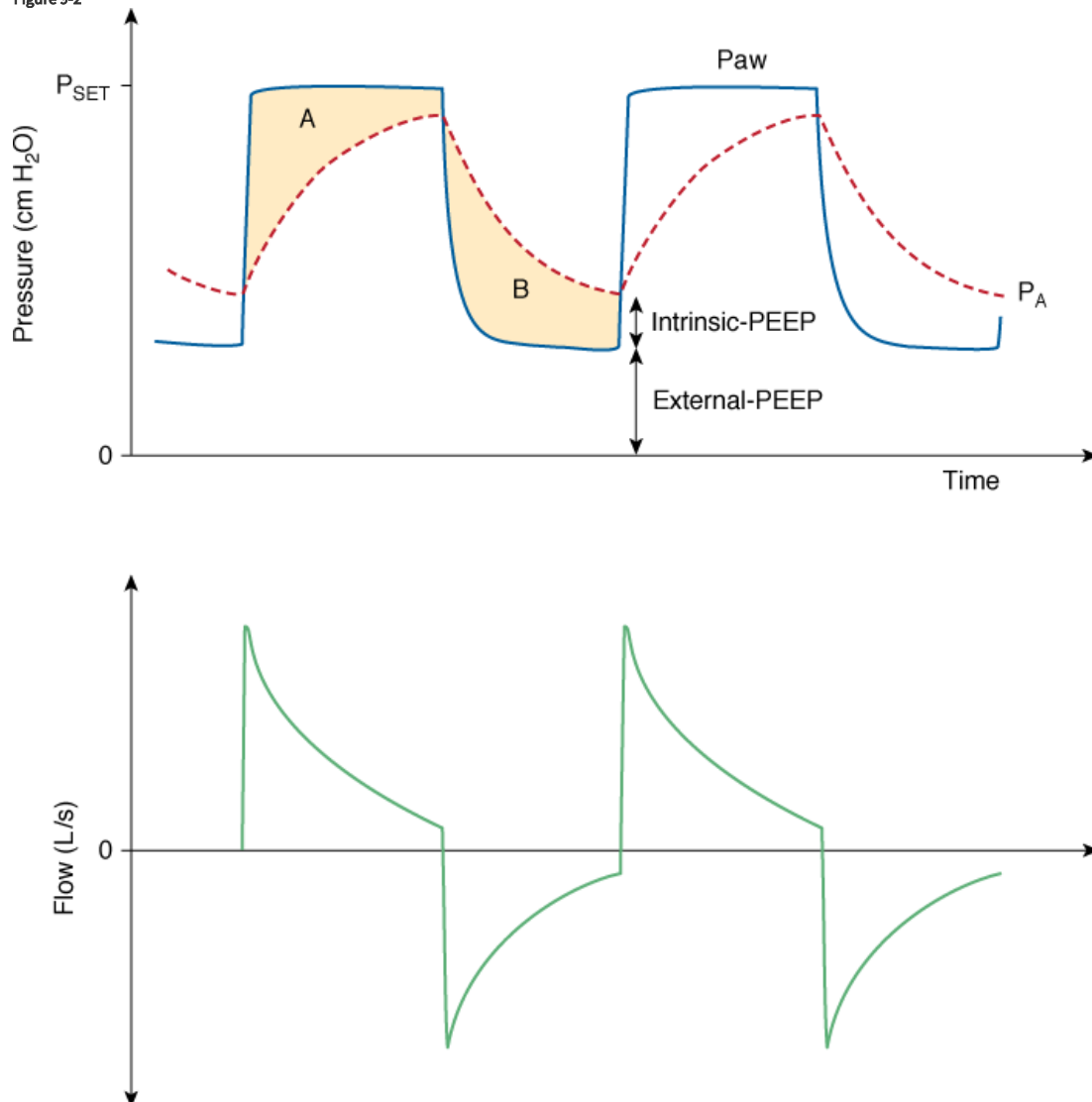
where R_E and R_I are mean expiratory and mean inspiratory resistances, respectively, and $\dot{V}_E$ is the minute ventilation.

Some practical conclusions can be drawn from this simple formulation. First, changing frequency alters mean alveolar pressure very little if inspiratory and expiratory resistances are similar. In this particular situation, mean airway pressures will reflect alveolar pressures consistently. Conversely, when expiratory resistance exceeds inspiratory resistance, a frequent condition in chronic obstructive lung disease, mean airway pressures can seriously underestimate mean alveolar pressures, especially when minute ventilation is high. Under such conditions, variations in frequency do influence mean alveolar pressure, and physicians easily could overlook the hemodynamic consequences of a ventilator setting.

Figure 9-2 illustrates some additional features related to alveolar and airway pressures during PCV. Alveolar pressure (P_A) can rise no higher than P_{SET} , the pressure to which it equilibrates at end inspiration when sufficient inspiratory time is provided (in Fig. 9-2 there is no equilibrium). Peak P_A falls rather than rises with increasing frequency. One also can grasp from the figure that intrinsic PEEP increases with

increasing frequency, reducing the *effective* ventilating pressure ($P_{\text{SET}} - \text{total PEEP}$) and consequently reducing tidal volume. As frequency increases further, P_A oscillations decline to the point that $\bar{P}_A$ sets the upper bounding value for intrinsic PEEP.

Figure 9-2



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Airway opening pressure (P_{aw}), alveolar pressure (P_A , red dashed line) and flow during pressure-controlled ventilation. Intrinsic-PEEP presents a backpressure that opposes the applied pressure (P_{SET}) and reduces the effective ventilating pressure, which is equal to $P_{\text{SET}} - \text{total-PEEP}$ (total-PEEP = intrinsic-PEEP + extrinsic-PEEP). Areas A and B represent the pressure-time product dissipated during inspiration and exhalation, respectively. These areas are proportional to mean inspiratory and expiratory resistances.

Under conditions of passive inflation, mean airway pressure reflects the average distending pressure of the respiratory system. Understandably, therefore, mean airway pressure has been associated with two beneficial physiologic effects (ventilation and oxygenation) and three potentially noxious effects (hemodynamic compromise, fluid retention, and barotrauma). These effects, however, are nonlinear, and there are exceptions. For instance, as discussed later, the relationship between $\bar{P}_{aw}$ and oxygenation is very dependent on the extent of pressure-volume hysteresis of the lung, which is greatly affected by lung disease. The greater the hysteresis, the greater is the dependency of oxygenation on PEEP and previous lung history (i.e., the maximum alveolar pressure achieved in previous breaths), and the looser is the correlation between the current $\bar{P}_{aw}$ and oxygenation. In situations of negligible lung hysteresis, for instance, during partial liquid ventilation, the correlation between $\bar{P}_{aw}$ and oxygenation is straight forward.³⁰ The complex relationship between $\bar{P}_{aw}$ and oxygenation is further elaborated in the short discussion of IRV below.

Mean airway pressure is also a measurable correlate of the back pressure for venous return. Raising $\bar{P}_{aw}$ during passive ventilation increases both lung and chest volumes by similar amounts. Lung expansion tends to increase right-ventricular afterload, which is already high in many

patients with acute respiratory failure. More important, the increase in intrapleural pressure may raise right-atrial pressure, often impeding venous return. Rising back pressure can have clinical consequences in patients with impaired systemic venous tone and reduced tissue turgor (e.g., consequent to sedation or paralysis). Sodium and water retention also tends to correlate with the magnitude of $\bar{P}_{aw}$. Although high $\bar{P}_{aw}$ may not be directly implicated in the *generation* of barotrauma (barotrauma seems to be mainly associated with high inspiratory plateau pressures or high inspiratory driving pressures³¹), a high level of $\bar{P}_{aw}$ may *exacerbate* damage or *accentuate* gas leakage through rents in the alveolar tissues, thereby bringing barotrauma to clinical attention.

Output Variables of Pressure-Controlled Ventilation

As discussed earlier, the major “output” variables of **PCV** are tidal volume, intrinsic PEEP, minute ventilation, alveolar ventilation, and inspiratory flow.

Tidal Volume

When maximal airway pressure is preset, the tidal volume actually delivered varies with several key variables: the pressure gradient existing between the airway opening and the alveolus at the onset of inflation, the resistance to airflow, the compliance of the respiratory system, and the time available for inspiration. Theoretically, inspiratory time should be longer than the three time constants of the respiratory system to allow near-complete (>95%) lung filling, thus maximizing delivered tidal volume. This scenario would guarantee that alveolar pressures are in equilibrium with airway pressures at the end of inspiration. In adult patients with orotracheal intubation, equilibration usually requires 1.0 to 1.5 seconds of inspiration. In the presence of severe obstructive disease, this value can be as high as 2 to 4 seconds, whereas in patients with reduced compliance (acute respiratory distress syndrome [ARDS]) this value can be as short as 0.8 to 1.0 second. To improve synchrony during *assisted PCV*, sometimes it is necessary to match the spontaneous inspiratory time of the patient. In this case, inspiratory time rarely should exceed 1.2 seconds, usually resulting in incomplete lung filling but promoting comfort.

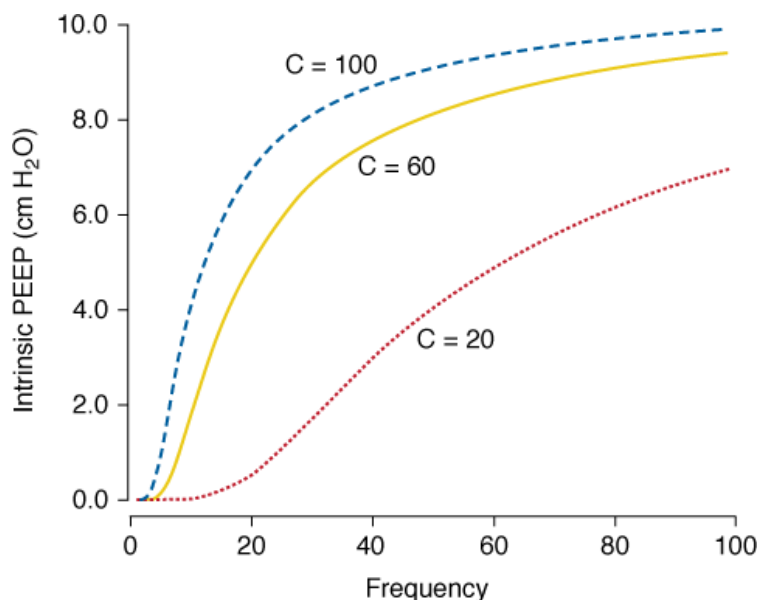
Any residual end-expiratory pressure (intrinsic PEEP) detracts from the total pressure difference available to accomplish ventilation (*driving pressure*). Therefore, incomplete lung emptying is also an important factor affecting the delivered tidal volume. To allow near-complete lung emptying, expiratory time should be longer than three expiratory time constants. The aggravating factor is that expiratory resistance is usually higher than inspiratory resistance, implying that the expiratory time constants are longer. An expiratory time shorter than 1.5 seconds in an adult patient should be considered a potential source of intrinsic PEEP, reducing the potential for tidal volume delivery.

Such relationships are also valid during low tidal ventilation (used for lung protection, often combined with tidal volumes of approximately 6 mL/kg or lower), which means that incomplete lung filling or emptying are common at respiratory rates above 25 to 30 breaths/min. Because the *absolute* values of intrinsic PEEP, however, in this setting tend to be negligible (from a clinical perspective),³² clinicians have used **PCV** with respiratory rates up to 40 breaths/min without great concern about intrinsic PEEP.

Intrinsic PEEP

Intrinsic PEEP (auto-PEEP) that results from dynamic hyperinflation is a complex function of the input parameters of **PCV** in conjunction with the impedance characteristics of the respiratory system.³³ The general principles affecting it can be summarized as follows: (a) higher-frequency, longer T_I/T_{TOT} ratio, and higher P_{SET} tend to increase intrinsic PEEP; (b) increments in T_I/T_{TOT} cause a monotonic increase in intrinsic PEEP from external PEEP up to P_{SET} ; (c) pure increments in frequency also cause an increase in intrinsic PEEP but with a saturation effect that limits intrinsic PEEP to half (approximately) the range between external PEEP and P_{SET} , which arises because as frequency rises, inspiratory time is curtailed, preventing equilibration between applied airway and alveolar pressures, keeping maximum PA well below P_{SET} (Fig. 9-3); and (d) the higher the compliance and the higher the expiratory resistance, the higher is the intrinsic PEEP for the same input parameters. Figure 9-3 illustrates some of these relationships.

Figure 9-3



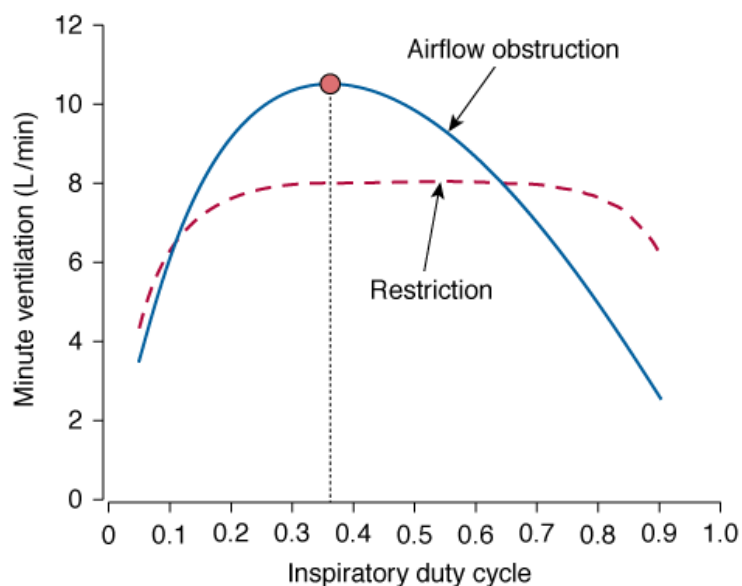
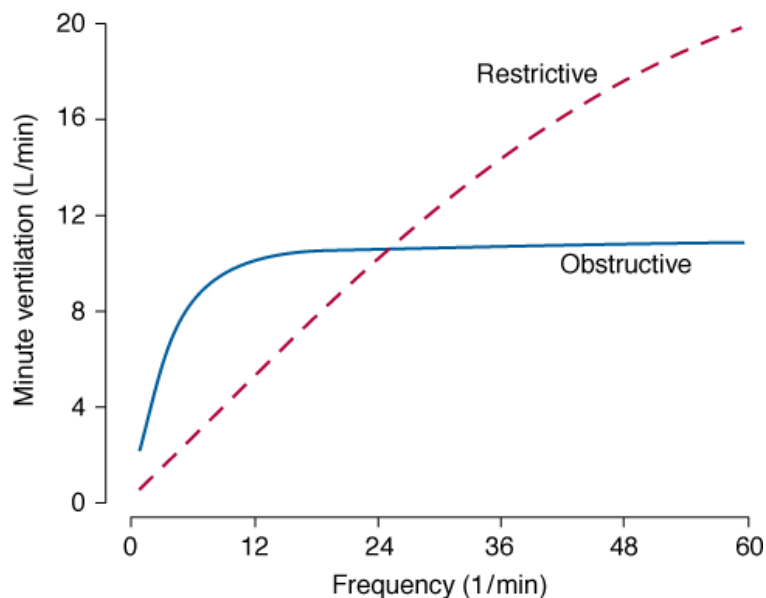
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Effect of frequency on intrinsic-PEEP generation during PCV. Three impedance combinations with moderate airflow obstruction and varying respiratory system compliance (expressed in mL/cm H₂O). A decrease in compliance causes both a reduction in intrinsic PEEP and alteration in the curvature of the intrinsic PEEP-to-frequency relationship. Note that curves converge to an asymptote at 10 cm H₂O, which corresponds to approximately half P_{SET}. *Simulated conditions:* P_{SET} = 20 cm H₂O above PEEP; T_I/T_{TOT} = 0.4; R_E = 25 cm H₂O/L/s. (Adapted from Marini JJ, Crooke PS 3rd. A general mathematical model for respiratory dynamics relevant to the clinical setting. *Am Rev Respir Dis*. 1993;147(1):14–24).

Minute Ventilation

The relationships just described are responsible for important and nonintuitive effects on minute ventilation (Figs. 9-4 and 9-5). When frequency increases at constant values for P_{SET} and T_I/T_{TOT}, the durations of inspiration and expiration both decrease, and intrinsic PEEP rises. As tidal volume falls, minute ventilation exponentially approaches an upper bound mainly determined by resistance.³³ In obstructed adult patients breathing at respiratory rates above 40 breaths/min, changes in compliance have a minor influence on minute ventilation generation, because most of the *driving pressure* is spent in overcoming resistive losses. For instance, a decrease in compliance from 100 to 50 mL/cm H₂O will produce a maximum fall in minute ventilation of 10%. This situation resembles a condition commonly observed during high-frequency oscillation,^{34,35} when the distribution of ventilation is mainly driven by the distribution of regional resistances, rather than regional compliances.

Figure 9-4

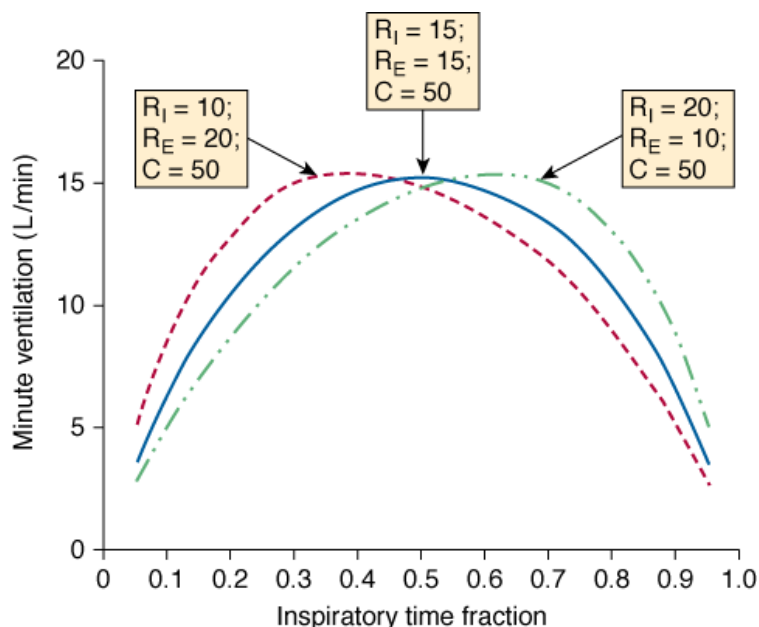


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Upper plot. Effect of frequency on minute ventilation during PCV. An increase in frequency causes tidal volume to fall so that minute ventilation approaches a mathematically defined plateau value mainly determined by the applied pressure and resistance. The approach is more gradual for a restrictive ventilatory defect with low compliance. **Bottom plot.** Unlike minute ventilation in the obstructed condition, for which a distinctly optimal T_I/T_{TOT} is evident (red dot), minute ventilation in the restricted condition remains essentially unaffected by changes in this parameter. **Simulated conditions:** $P_{SET} = 20$ cm H₂O above PEEP; $T_I/T_{TOT} = 0.4$; restrictive, $C = 20$ mL/cm H₂O, $R_I = R_E = 10$ cm H₂O/L/sec; obstructed, $C = 100$ mL/cm H₂O, $R_I = 15$, $R_E = 45$ cm H₂O/L/sec. (Adapted from Marini JJ, Crooke PS 3rd. A general mathematical model for respiratory dynamics relevant to the clinical setting. *Am Rev Respir Dis*. 1993;147(1):14-24.)

Figure 9-5



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Effect of inspiratory time fraction (T_I/T_{TOT}) on minute ventilation according to the balance between inspiratory and expiratory resistances (R_I and R_E , respectively). As is true for tidal volume, the apogee of the curve is reached at $T_I/T_{TOT} = 0.5$, provided that $R_I = R_E$. Note that optimum T_I/T_{TOT} is shifted leftward, when $R_E > R_I$, exemplifying a common situation in clinical practice. (Adapted from Marini JJ, Crooke PS, 3rd, Truitt JD. Determinants and limits of pressure-preset ventilation: a mathematical model of pressure control. *J Appl Physiol*. 1989;67(3):1081–1092.)

Figure 9-4 suggests that in patients with moderate to severe airway obstruction (resistance ≥ 25 cm H₂O/L/s), the benefits of increasing respiratory rate above 12 breaths/min is negligible (this is an important hint when ventilating patients with asthma). The situation changes considerably, however, under conditions of low compliance (≤ 25 mL/cm H₂O): any increase in frequency becomes an effective mechanism to increase minute ventilation (see Fig. 9-4, top).

Important differences between obstructed and restricted patients also can be seen in the relationship between minute ventilation and duty cycle. Minute ventilation becomes very sensitive to changes in duty cycle for obstructed patients but not for restrictive patients. As shown in Figure 9-5, if one wants to maximize minute ventilation for a given P_{SET} (a common target during intensive care of patients with asthma), T_I/T_{TOT} has to be titrated according to the ratio between inspiratory and expiratory resistances. For patients with fixed obstruction and equivalent values for inspiratory and expiratory resistances (as sometimes occurs in asthmatic patients), the ideal T_I/T_{TOT} for maximizing minute ventilation is 0.5.

Alveolar Ventilation

Over the lower frequency range, increasing frequency tends to improve alveolar as well as total ventilation, decreasing partial pressure of arterial carbon dioxide (P_{aCO_2}); however, the same is not necessarily true when high frequencies are used. As frequency rises at a fixed, noninverse T_I/T_{TOT} , inspiratory time is curtailed, preventing equilibration between applied airway and alveolar pressures.

From a practical standpoint, the ventilator itself becomes less able to generate the nominal pressure waveform, especially when flow impedance is low and *rising-time* is not optimal. As tidal volume declines, the wasted fraction of each breath (dead-space-to-*tidal-volume* ratio [V_D/V_T]) increases owing to the predominance of the series (“anatomic”) dead-space component. Under certain conditions, this increase in dead space actually may cause P_{aCO_2} to rise rather than fall with increasing frequency.³⁶ In practical terms, there is an important message for the bedside: For a given P_{SET} and T_I/T_{TOT} , increments in frequency may cause a decrease in P_{aCO_2} up to the point that tidal volume decreases by 25% to 30%. Beyond this limit, even when minute ventilation increases with frequency, it is likely that further increments in frequency will be counterproductive because of excessive amounts of dead space.

Because the principles just outlined are rooted in physics and mathematics, hypercapnia can be an unavoidable consequence of a pressure-targeted strategy for managing acute lung injury.

Inspiratory Flow

Decelerating inspiratory flow necessarily is observed during pressure-controlled breaths with rectilinear pressure waveforms—provided that there is no patient effort. Under such conditions, the theoretical maximum of flow associated with a square wave of pressure (P_{SET}) depends on inspiratory resistance (R_I) and end-expiratory alveolar pressure ($PEEP_{TOT}$), and is achieved at inflation onset:

$$Peak\ flow = \frac{P_{SET} - PEEP_{TOT}}{R_I} \quad (3)$$

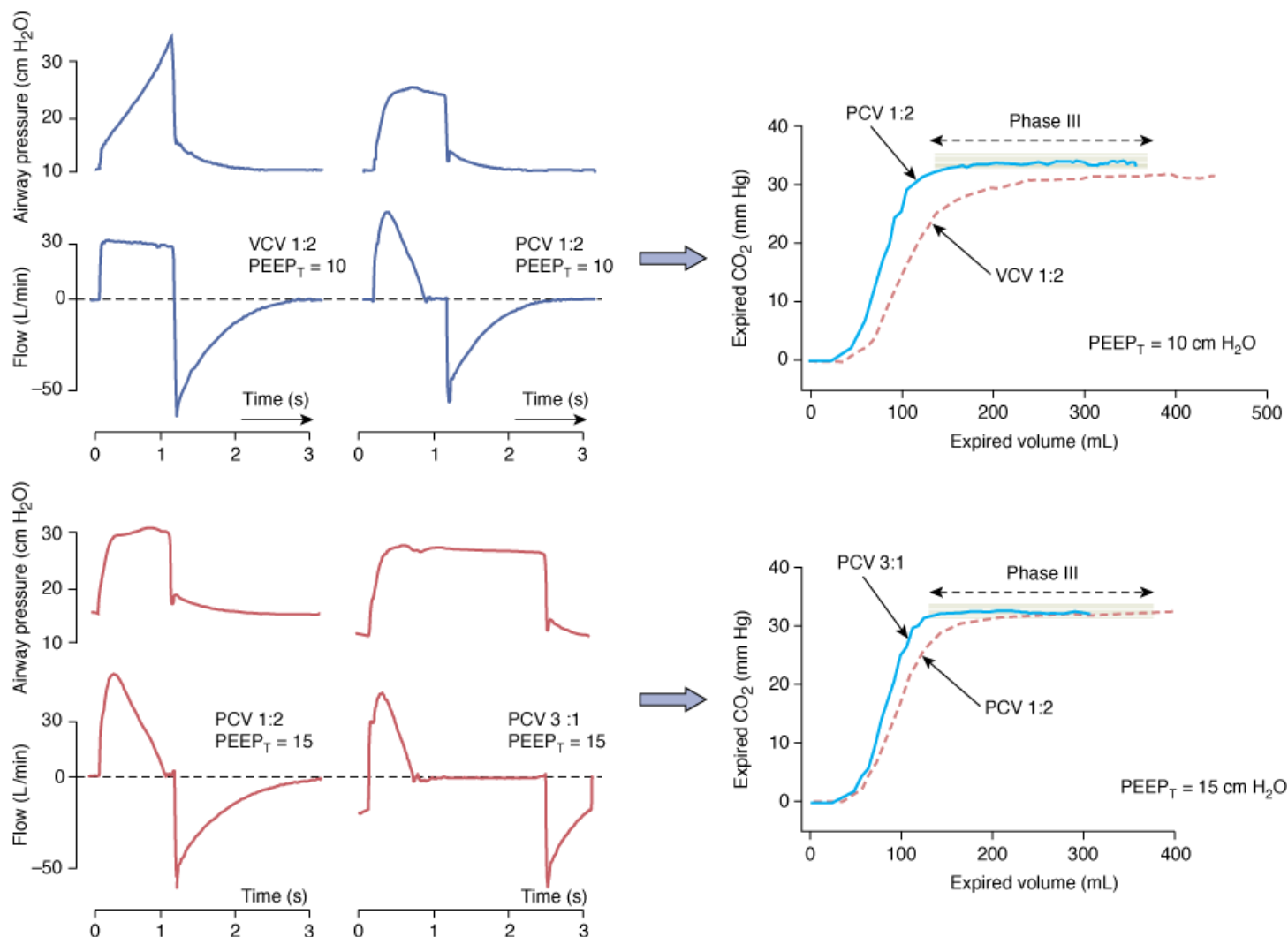
where $PEEP_{TOT}$ = intrinsic PEEP + external PEEP.

In practice, however, the abruptness of the rise to the nominal peak value is a set characteristic of the particular ventilator, which may be modulated by the *slope* or inspiratory *rise-time* adjustment. Under conditions of quiet breathing, a precipitous buildup to peak flow often is associated with some pressure overshoot, which may be annoying for monitoring purposes because of alarm triggering. Because most of this pressure overshoot represents pressure dissipation as frictional work across the endotracheal tube, it does not cause elevation of peak alveolar pressure and probably is *not* associated with any harm.

Conversely, under high flow demands (especially in large patients, or in patients using the helmet for noninvasive ventilation), a slow “attack rate” up to peak flow can cause some pressure undershoot or a slow ramp of pressure at the initial phase of the breath, forcing the patient to expend considerable effort and causing delaying filling of the lung.^{37,38} Because of limitations in the hardware controlling system, the flow performance of most ventilators, especially at the first 300 milliseconds of inspiration, tends to be poor when P_{SET} is less than 10 cm H₂O. Under such conditions, maximizing rise time or slightly increasing P_{SET} (in association with some procedure to avoid too prolonged T_I , such as shortening the set T_I during assisted PCV, or increasing the cycling-off criterion during PSV³⁹) can be very helpful.

The decelerating-flow pattern found in PCV usually improves the distribution of ventilation and limits the end-inspiratory gradient of regional pressures among units with heterogeneous time constants. When inspiratory time is long enough, inspiratory flow may decrease down to zero before exhalation, a phenomenon that further favors redistribution of air or pressure among heterogeneous units and collateral ventilation.^{40,41} The consequences of such a flow profile are reflected mainly in the CO₂ eliminated per breath (Fig. 9-6).⁴² The longer the inspiratory time, the more effective is the clearance of CO₂, although most of the benefit may be seen already when T_I slightly exceeds the point of zero flow (Fig. 9-6). This topic is discussed further in “The Controversy on Optimal Distribution of Ventilation” below.

Figure 9-6



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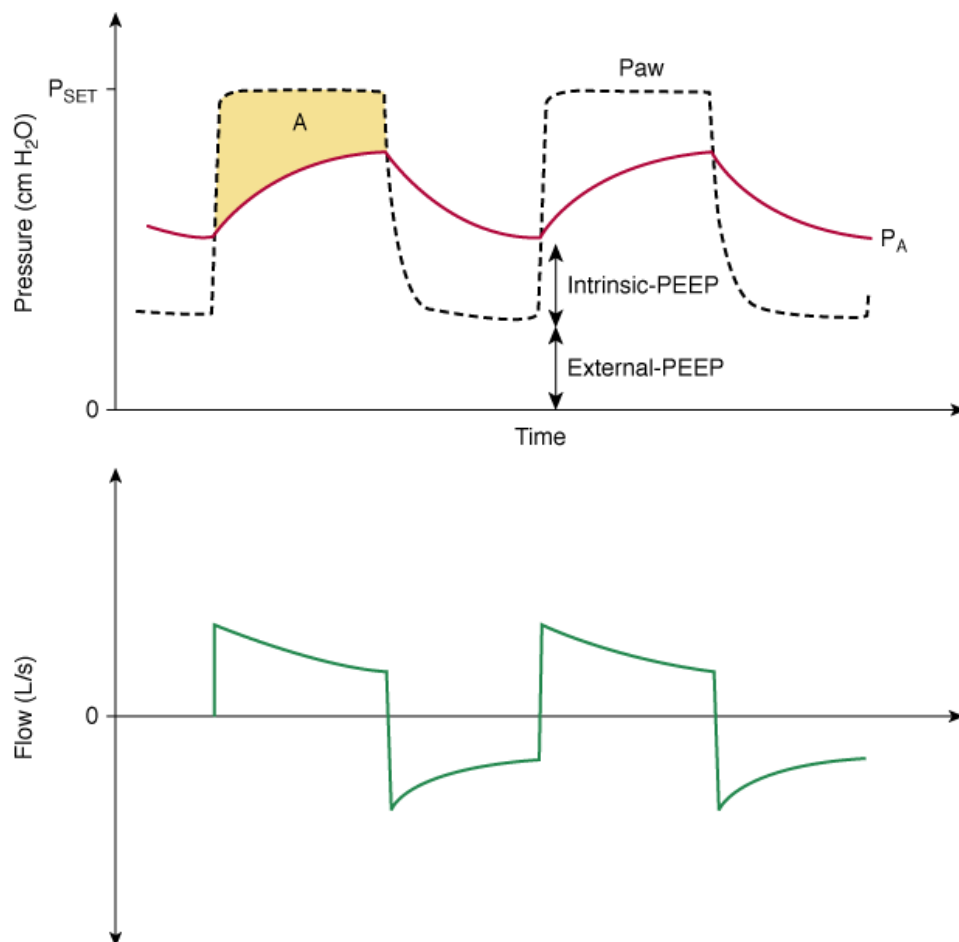
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Effects of inspiratory flow profile on CO_2 elimination per breath (CO_2 single-breath tests obtained in a mainstream volumetric capnograph).

Top. The change from flow-controlled, volume-cycled ventilation (VCV) to PCV, while keeping the same inspiratory time, resulted in more efficient elimination of CO_2 per breath, reflected by the large area under the curve of CO_2 versus exhaled-volume (especially in the first 200 mL, and consequently reflecting a lower dead space). Note that inspiratory flow decayed to zero before exhalation and that the phase III slope is almost flat during PCV, reflecting less heterogeneity among lung units. Arterial P_{CO_2} was the same (38 mm Hg) in both conditions, despite the lower tidal volume during PCV. *Bottom.* Changing PCV with an I:E ratio = 1:2 to PCV with an I:E ratio = 3:1 (when part of external PEEP was replaced by intrinsic PEEP) resulted in a further increase in the area under the curve of CO_2 versus exhaled volume. The benefit, however, was much less evident than in the *top panel*.

Figures 9-7 and 9-8 illustrate the effects of resistance and compliance on flow profile. Because of the consequent increase in time constant, increments in resistance tend to produce a less decelerating flow pattern (presenting reduced peak flow, with more squared appearance) and rendering tidal volume very sensitive to reductions in inspiratory time. In contrast, decrements in compliance tend to accelerate flow decay. It is obvious, therefore, that restrictive patients tolerate a shorter inspiratory time without marked consequences to their tidal volume.

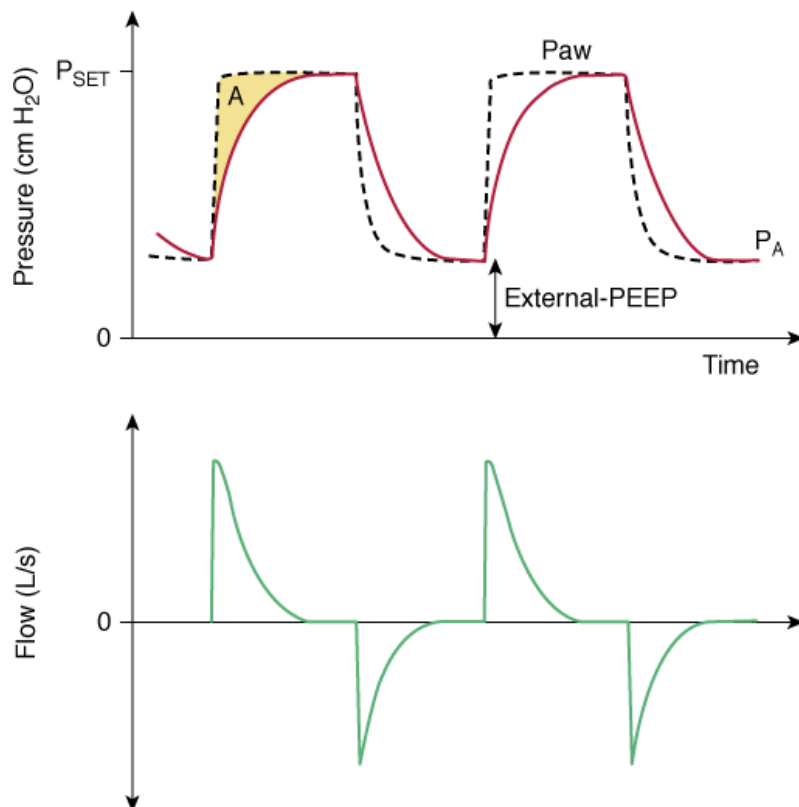
Figure 9-7



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Typical tracings in an obstructed patient receiving PCV. Note that the flow pattern resembles a square waveform, mimicking *flow-controlled, volume-cycled* ventilation. Intrinsic PEEP is evident and end-inspiratory alveolar pressures (in red) are much lower than end-inspiratory airway pressures (black dashed line).

Figure 9-8



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Typical tracings observed in a restricted patient receiving PCV. Note that inspiratory flow decelerates quickly, achieving zero-flow conditions well before the end of inspiration (generating a small area A; compare with Fig. 9-7). There is no intrinsic PEEP and end-inspiratory alveolar pressures (in red) equilibrate with end-inspiratory airway pressures (black dashed line).

Physiologic Effects of Pressure-Controlled Ventilation

Advantages of Controlling Airway Pressures

A major feature of PCV is that peak alveolar pressure cannot rise any higher than P_{SET} . In critical situations, when one attempts to minimize ventilator-induced lung injury, this aspect of PCV may be convenient. Specifying the maximum achievable alveolar pressure, however, does not cap the upper limit for *transalveolar* pressure (equivalent to transpulmonary pressure during an end-inspiratory pause) unless the patient's own breathing efforts also have been silenced (see Fig. 9-1). As suggested by many studies, transalveolar pressure, rather than alveolar pressure, is the key determinant of ventilator-induced lung injury and barotrauma.^{43,44}

In the absence of patient efforts, keeping peak alveolar pressures within a safe range makes sense. Under these circumstances, controlling airway pressure effectively controls maximal alveolar pressure. Obviously, the same peak alveolar pressure always can be achieved by flow-controlled, volume-cycled ventilation, although more bedside adjustments are necessary. The subtle difference here is that by selecting *pressure* as the controlling parameter, the physician better defines the priority of his or her strategy (i.e., to minimize peak alveolar pressure at the expense of minute ventilation and possible hypercapnia) and probably minimizes violations to the target during ongoing tidal ventilation (see the section [The Controversy on Optimal Distribution of Ventilation](#)).

When thinking in peak alveolar pressures as opposed to peak airway pressures, it is important to stress some important aspects of PCV. When inspiratory time is brief, end-inspiratory airway and alveolar pressures fail to equilibrate, so the maximal alveolar pressure is considerably *less* than the set value. This is reflected by persistent end-inspiratory flow, which frequently occurs in obstructed patients.

For a given inspiratory time and peak airway pressure, however, PCV is the waveform that applies the greatest cumulative pressure to the respiratory system.⁴⁵ Therefore, for the *same* peak airway pressure—and provided that inspiratory time is long enough—PCV generates a higher peak alveolar pressure and a higher tidal volume than VCV (flow-controlled, volume-cycled breaths delivered with a square-wave profile). Such characteristics of PCV are advantageous during laryngeal mask ventilation: for the same delivered tidal volume, PCV generates

lower peak laryngeal pressures during inspiration (approximately 3 to 4 cm H₂O lower than during VCV), resulting in less inadvertent gastric insufflation,⁴⁶ and keeps peak airway pressures at a safer distance from the threshold leak pressures.⁴⁷

The consequences of PCV on $\bar{P}_{aw}$ were discussed earlier. It is an important parameter for evaluating the hemodynamic consequences of PCV. Unlike in flow-controlled VCV, $\bar{P}_{aw}$ relates linearly to P_{SET} and T_I/T_{TOT} (see Eq. 1). As its defining equation indicates, $\bar{P}_{aw}$ is unaffected by changes of respiratory system impedance and frequency. Provided that changes in inspiratory and expiratory resistance are roughly balanced, mean alveolar pressures follow mean airway pressures very consistently. Consequently, the impact of adjustments in $\bar{P}_{aw}$ on $\bar{P}_A$ can be predicted much more easily during PCV than during VCV.

Such a straightforward relationship between $\bar{P}_{aw}$ and the input parameters of PCV may be convenient during short-term procedures such as recruiting maneuvers. By adjusting PEEP, P_{SET} , and T_I/T_{TOT} , one can easily predict the generated $\bar{P}_{aw}$ (or $\bar{P}_A$) and hence the hemodynamic consequences. Recent studies demonstrate the relative safety of recruitment maneuvers using PCV for 1 to 2 minutes,^{15–17} which achieved equivalent or superior efficacy as sustained pressure maneuvers (continuous positive airway pressure [CPAP]) adjusted to equivalent P_{SET} .^{15,48–51} Because the motor of effective recruitment is the surpassing of threshold opening pressures of terminal airspaces,⁵² applied long enough to overcome the forces of viscosity and adhesion,⁵³ cyclic pressurizations with PCV provide an interesting alternative. At the same time that repeated waves of inspiratory pressure promote progressive recruitment, repeated relief of pressure during exhalation minimizes impairment of venous return. Theoretically, one should adjust P_{SET} above the threshold opening pressures, adjust PEEP above the closing pressures, and ensure that inspiratory time is long enough to favor slow, sequential stepwise recruitment of clumps of alveolar units.⁵² As suggested by Neumann et al,⁵⁴ inspiratory times exceeding 0.6 second (ideally closer to 3 seconds)—for instance, obtained during PCV with a frequency of 8 to 10 breaths/min and an I:E ratio of 1:1—would be enough to maximize the potential for recruitment at a certain P_{SET} . The great appeal of such PCV maneuvers is their reasonable hemodynamic tolerance: because the $\bar{P}_{aw}$ generated during a PCV maneuver is substantially less than the $\bar{P}_{aw}$ generated during a CPAP maneuver (at equivalent P_{SET}), the hemodynamic consequences are less pronounced.^{48,50,51,55} Preceding volume expansion with colloids further improves hemodynamic tolerance of the maneuver.⁵⁰

Some investigators, such as Schreiter et al,⁵⁶ have applied PCV recruiting maneuvers over much shorter periods (approximately 10 to 15 seconds) but reached higher inspiratory pressures (50 to 80 cm H₂O) and reported good success in patients with severe chest trauma.

Comparative data of the relative efficacy and safety of shorter, more intensive application of recruiting pressures^{57–60} versus longer (1 to 2 minutes) application of less intensive (45 to 60 cm H₂O) recruiting pressures^{15,17,61–63} do not yet exist.

When some active patient effort is present during PCV, the primary control of airway pressures (instead of flow) might offer the potential benefit of introducing some variability in the flow or tidal volume profiles among breaths. By design, an assisted PCV breath provides more freedom (lower machine impedance) for the intrinsic variability of a patient's respiratory motor output to manifest. Besides the likely improvement in comfort, recent research suggest that some random variation (within certain limits) in the effective driving pressure applied to the respiratory system may bring additional benefits in terms of oxygenation, mechanics, and surfactant function.^{64–69} The benefits of such extra freedom, however, have to be tempered in conditions where lung protection is a priority, especially when patient effort becomes too vigorous (see Fig. 9-1).

As with any time-cycled mode of ventilation, PCV invites dyssynchrony when the patient breathes spontaneously. The implications of slow *rising-time* settings were discussed earlier. It is important to think about the other end of the inflation period, however, when the airway continues to be pressurized to the nominal value until the set inspiratory time has elapsed. As with flow-controlled VCV, the patient may attempt to cycle to expiration before the ventilator completes its pressurization cycle. One could imagine, however, that unlike the situation with flow-controlled ventilation, patient effort never could force airway pressure higher than the preset value during PCV. Unfortunately, this is not always true (see the section [Activation of Exhalation Valve During Pressure-Controlled Ventilation: Airway Pressure Release Ventilation](#)) because of hardware limitations in some ventilators.

Another feature related to dyssynchrony is the fact that by fixing T_I , as required with many commercial ventilators, one allows T_I/T_{TOT} to vary with frequency whenever the patient retains control of the cycling rhythm. As frequency increases, T_{TOT} decreases and T_I/T_{TOT} tends to increase, often provoking dynamic overinflation and generation of intrinsic PEEP. Even with ventilators in which the direct input is “duty cycle,” most of them calculate the inspiratory time based on the *set* frequency rather than on *measured* frequency. Therefore, the resulting T_I/T_{TOT} can be very different from the originally set T_I/T_{TOT} .

The Controversy on Optimal Distribution of Ventilation

Several studies have attempted to demonstrate a definitive advantage of PCV over VCV (squared flow pattern) in terms of ventilation distribution, oxygenation, hemodynamics, lung injury, and patient outcome. At first glance, results seem inconclusive. By separating the studies according to key ventilator parameters measured, however, some consistency emerges.

First, it is important to distinguish studies that compare PCV-IRV from studies comparing PCV with normal I:E ratio against other modes. As discussed within this section, the effects of IRV are complex and depend on intrinsic PEEP generation. Thus, during IRV, the choice of the controller (*flow* versus *pressure*) is just a minor issue inside a complex and broader picture. Second, it is important to observe whether all the remaining variables (I:E ratio, tidal volume, intrinsic PEEP, total PEEP, frequency, and O₂ fraction) were kept constant during the comparison and whether they were consistent. For example, studies displaying different plateau pressures or tidal volumes during PCV and VCV can hardly be considered to be perfectly matched if one is interested in the effects on oxygenation⁷⁰ or lung overdistension.⁷¹ Thus, they should not be used to draw important conclusions on this controversy.

Keeping such boundaries in mind and by considering only the effects of a pure change in the type of controller (from squared *flow control* to squared *pressure control*, while keeping similar values of I:E ratio, tidal volume, frequency, total PEEP, and plateau pressure), most studies have demonstrated that:

The resulting decelerating flow during PCV decreases peak airway pressures but increases $\bar{P}_{aw}$.^{71–79} Such reduction is helpful during laryngeal mask ventilation,⁴⁶ avoiding high peak pressures at the esophageal entrance and avoiding gastric insufflation.⁸⁰ PaCO₂ and V_D/V_T decrease slightly (Δ PaCO₂ approximately 2 to 4 mm Hg), with modest clinical significance,^{72,75–77,79} in accordance with the example shown in Figure 9-6. PCV seems to favor ventilation of units with slow time constants, increasing also the *mean distribution time* (defined as the time available for gas distribution and diffusion during inspiration^{41,42}). The fast delivery of flow during early inspiration is probably responsible for this effect.

Such selective benefit in terms of ventilation distribution has been explored theoretically and clinically in obstructed patients.^{81–83} PCV is certainly an interesting option in conditions of severe CO₂ retention. Theoretically, however, if obstruction is too severe, the decelerating flow waveform of PCV may become relatively “squared” (see Fig. 9-7), and some benefits reported for decelerating *flow-controlled* VCV⁸⁴ (which necessarily achieves zero flow at end-inspiration) may not be directly translated to PCV.

Minor, if any, changes in dynamic (within the breath) or static lung aeration (end-expiratory or end-inspiratory pause) are observed by computed tomography.^{71,74,78} or by mathematical modeling.⁸⁵ Regional lung strain measured by computed tomography was also equivalent.⁸⁶ Peak alveolar pressure distribution across lung units, however, seems to be more favorable during PCV, guaranteeing a lower exposure to high pressures in more diseased areas of the lung.⁸⁵

Minor changes are seen in the partial pressure of arterial oxygen (PaO₂),^{72,74,84} slightly favoring PCV.^{75,79,87,88} Numerous studies have evaluated the advantages of PCV during anesthesia and one-lung ventilation (during thoracic surgery)^{89,90} or during laparoscopic exploration of abdominal cavity,^{91–93} including in morbidly obese patients.^{94–96} All showed equivalence or minor advantages of questionable clinical relevance.

No hemodynamic impairment is seen on switching from VCV to PCV, provided that the increments in $\bar{P}_{aw}$ are moderate (usually the case when using I:E<1:1).

No differences are seen in barotrauma or patient outcome. Studies suggesting benefit of PCV had imbalanced randomization,⁸⁸ whereas some studies suggesting benefit of VCV were poorly controlled (including crossovers and rescue procedures).^{97–99}

Taken together, the evidence suggests a clear physiologic difference between the two modes of flow delivery, although the clinical implications are modest. At the bedside, the most consistent effects are a reduction of peak inspiratory airway pressures, increase in $\bar{P}_{aw}$, and small reduction in PaCO₂. Because of the increment in $\bar{P}_{aw}$, PaO₂ tends to increase with PCV, but this depends on PEEP and plateau pressure in the particular case.

Some concerns are raised repeatedly about the potential generation of greater shear forces at the beginning of inspiration during decelerating-flow patterns. It is suggested that the squared-flow waveform might distribute stress more evenly throughout inspiration, theoretically decreasing VILI. This notion, however, is just theoretical speculation, valid in only extreme conditions. The isolated effects of

high inspiratory flow rates on VILI^{100–109} have been investigated in recent years. Most studies suggest a deleterious effect,^{100,104,106–109} although a few suggest a neutral or protective effect.^{102,103,105} The deleterious effects, however, could be demonstrated only in association with tidal volumes above 20 mL/kg, and in conditions where PCV caused an approximate three fold increment in inspiratory flow, as compared to VCV.

In the context of protective ventilation, the peak flow rates observed during PCV are much lower, usually within the same range as those achieved during VCV.²⁵ When the driving pressure set on the ventilator is within the 15 to 20 cm H₂O range, the resistance across the endotracheal tube and connectors is usually enough to avoid peak flow rates above 1.5 L/s. Thus, consistent differences in VILI will hardly be linked to PCV as opposed to VCV, at least during controlled ventilation and during ventilation with small tidal volume (≤ 6 mL/kg). Isolated short-term changes in the controller, from flow to pressure control, could probably induce mild perturbations in the effective stress applied on the parenchyma.^{85,86} But when compared with the potential for iatrogenic damage associated with other parameters set on the ventilator, such as the choice of tidal volumes or resulting driving pressures (plateau pressure minus PEEP during controlled ventilation), the controller option is most likely insignificant.

As discussed earlier, with adequate feedback it is usually possible to obtain the same ventilator output with either a *flow controller* or a *pressure controller* under specific conditions. For instance, a flow-controlled breath with a decelerating-flow waveform can be generated by modern ventilators, mimicking the flow and airway pressure waveforms commonly observed during PCV^{79,110} and producing some of its physiologic effects. Because it is practically impossible to match the degree of deceleration with the impedance characteristics of different patients (e.g., more obstructed patients would present a less accentuated deceleration during PCV), peak airway pressures are frequently higher during such pressure-control “imitations.”^{110,111}

Considering the evidence presented above, we again suggest that the choice for PCV should be decided in terms of a global bedside strategy and not based on subtle physiologic differences. Minor differences between PCV and VCV may have an impact only when considering the associated clinical problems. For instance, in a patient requiring high PEEP levels, in whom PaCO₂ and alveolar dead space are also increased, and who thus requires high plateau pressure to achieve satisfactory ventilation, a change from VCV to PCV might be appropriate because (a) plateau pressures will rarely exceed the physician-adjusted P_{SET}; (b) any decrease in dead space afforded by decelerating flow may allow further reduction in tidal volumes and, thus, in P_{SET}; (c) an eventual rise in respiratory rate triggered by the patient will not be translated into increased plateau pressures despite the possible generation of intrinsic PEEP; and (d) paralyzing agents and sedation may thus be reduced.

Paradoxically, however, such advantages may be quickly nullified if spontaneous efforts become too strong. By decreasing pleural pressures, diaphragmatic activity may cause transpulmonary pressures to be excessive, despite rigid control of maximum alveolar pressures (capped at P_{SET}, see Fig. 9-1). In fact, during strong patient efforts, a *flow controller* avoids excessive transpulmonary pressure more efficiently than a *pressure controller*, because airway pressures follow changes in pleural pressure more closely, being transiently reduced during patient effort (there is no extra “demand” flow to maintain airway pressure). The result is an almost constant transpulmonary pressure. This important physiologic difference may explain some evidences pointing to a better outcome in neonates during *flow-controlled* VCV¹¹² versus pressure-limited ventilation (which is essentially similar to PCV working with an open exhalation valve; see below the section “[Variants of Pressure-Controlled Ventilation and Activation of Exhalation Valve](#)”). Irregular breathing patterns and strong spontaneous efforts are especially common in neonates; the eventual coupling of such efforts with a *pressure-controlled* breath can transiently promote excessive transpulmonary pressures.

This is why we strongly recommend the close monitoring of the extra tidal volume generated by a patient during assisted PCV. Even when P_{SET} is relatively low, it is important to infer the effective driving pressures applied to the respiratory system by observing the resulting tidal volume during synchronous efforts. The simple division of tidal volume by respiratory compliance provides a good estimate of the effective driving pressure (the driving pressure applied across the respiratory system [the lung plus chest wall in series]). It is common to observe patients with ARDS submitted to PCV with P_{SET} at 10 cm H₂O, and producing tidal volumes of 500 mL during synchronous efforts. If respiratory system compliance is 20 mL/cm H₂O, such a presentation means that the effective inspiratory driving pressure is equivalent to 25 cm H₂O. Failure to adhere to these principles can make PCV more dangerous than VCV.

Another point considered here is that peak-flow rates during PCV at low levels of P_{SET} may be equal to lower than during VCV, increasing work of breathing.^{25,113} This might appear quite surprising, considering the previous rationale that the *free* inspiratory demand-flow during

assisted PCV is an effective tool to reduce the work of breathing.¹¹⁴ One should note, however, that this rationale was developed in an era when high levels of P_{SET} were common in clinical practice. This topic is discussed in the section [Assisted Pressure-Controlled Ventilation](#).

Assisted versus Controlled (Time-Triggered) Ventilation

Most reasons to control ventilation can be classified as needs to regulate the ventilatory pattern, reverse life-threatening abnormalities in blood gases, reduce agitation and the O₂ cost of breathing, or prevent excessive excursions of the injured lung. Perhaps the most common indication is to provide ventilator support to a patient with marginal cardiorespiratory reserve in whom breathing efforts compromise comfort, gas exchange, or cardiac function. Reducing O₂ consumption in a patient with critical coronary ischemia, for example, may prove lifesaving, especially in the setting of circulatory shock or acute pulmonary edema. Silencing ventilatory effort also can improve arterial oxygenation and lung volume recruitment in agitated patients with ARDS,^{115–119} probably by reducing expiratory efforts that may reduce end-expiratory transpulmonary pressures. The silencing of vigorous patient efforts during the first few hours of protective ventilation may also help by decreasing lung inflammation,^{119,120} and reducing double triggering (“breath stacking”).¹²¹ The active control of minute volume to achieve normocapnia or to achieve transient hypocapnia, when reductions in cerebral blood volume and intracranial pressure are urgent priorities, are also common indications for controlled ventilation. Controlling ventilation greatly facilitates the monitoring of respiratory mechanics but in itself is seldom sufficient reason to undertake deep sedation or pharmacologic paralysis.

Apart from reducing O₂ consumption and preventing spontaneous efforts from interfering with intended patterns of *conventional* ventilation, establishing ventilatory control facilitates the therapeutic application of “nonphysiologic” breathing patterns. Interventions such as IRV,^{9,13} independent lung ventilation,¹²² continuous-flow ventilation,¹²³ and permissive hypercapnia^{1,3,4,124} usually require cessation of patient effort. In patients with acute lung injury, use of small tidal-volume ventilation is a priority, but this pattern is not easily tolerated by patients unless deep sedation, ventilator overriding,^{125,126} or complete paralysis is applied.¹²⁰ This is especially true when tidal volume is set at 4 to 6 mL/kg.^{125,127}

Sedation is commonly used to enhance comfort or inhibit excessive respiratory drive. Muscle relaxants, however, should be used sparingly because of their injurious potential. A better understanding of the sequelae of prolonged paralysis, especially when combined with sepsis and systemic inflammation,¹²⁸ high-dose corticosteroids and hyperglycemia,^{129–131} has forced physicians to minimize the magnitude and duration of paralysis in ventilated patients. Recent work suggests that severe diaphragmatic dysfunction occurs after only 2 to 3 days of controlled mechanical ventilation,^{132–134} independent of the use of paralyzing agents, with loss of more than 40% of power generation. It is likely that paralyzing agents magnify or trigger such effects¹³⁵ by blocking protective mechanism against atrophy. Independent of its precise mechanism, critical illness polyneuropathy increases the duration of mechanical ventilation significantly.¹³⁶

Recent evidence suggests that calm assisted ventilation, with efforts just sufficient to trigger the ventilator, can avoid excessive muscle wasting.¹³³ Because spontaneous efforts also can improve oxygenation during pressure-controlled breaths,^{65,137–139} probably secondary to redistribution of blood flow to nondependent, well-ventilated lung areas, it is tempting to conclude that some level of spontaneous effort always should be promoted in patients receiving PCV. Such approach consistently reduces sedation requirements.¹³⁹ An anticipated drawback of such a strategy, however, is an undesirable increase in I:E ratio when the patient respiratory rate is much higher than the set respiratory rate. Another problem is the generation of unpredictably high transpulmonary pressures during strong and synchronous effort by the patient (see [Fig. 9-1](#)). In both circumstances, however, deeper sedation or partial paralysis could control the situation easily. By keeping the spontaneous respiratory rate close to the set rate, one easily can avoid changes in the I:E ratio. Also, by keeping the exhaled tidal volume close to a value associated to low driving pressures during totally controlled breaths (see the section [The Controversy on Optimal Distribution of Ventilation](#)), one may ensure that a patient’s spontaneous effort does not cause significant increments in inspiratory transpulmonary pressures.

Variants of Pressure-Controlled Ventilation and Activation of Exhalation Valve

As discussed earlier, the variants of traditional PCV^{10,75} include assisted PCV,^{25,114,140} PSV,^{141,142} and variants of APRV.^{143–147}

Assisted Pressure-Controlled Ventilation

Assisted PCV is similar to PSV in that the breaths can be triggered by the patient (physicians set the sensitivity and triggering mechanism). The difference is that cycling off is determined by time instead of flow. Another obvious difference is that assisted PCV possesses the same backup mechanism available with traditional assisted flow-controlled breaths: breaths can be triggered by patient effort or by elapsed expiratory time, whichever occurs first.

Compared with assisted VCV at equivalent tidal volumes, assisted PCV decreases peak airway pressures. Although irrelevant during invasive ventilation, this characteristic is relevant during noninvasive ventilation, minimizing leaks through the mask^{80,148} and avoiding uncomfortably high pressures in the upper airways.^{46,149} When comparing the level of patient assistance during invasive ventilation, the characteristic decelerating flow (*on demand*) provided by PCV seems to reduce workload more efficiently than most flow-controlled, volume-cycled breaths.^{114,140} The reduction in inspiratory muscle load is especially prominent during moderate to high tidal volume (8 to 10 mL/kg) ventilation or use of a low inspiratory flow during VCV (<0.7 L/s). It is important to stress, however, that with use of low tidal volume (≤ 6 mL/kg), this assumption is not always valid.^{25,113,125} Because physicians have to set low values of P_{SET} during *pressure-controlled* breaths (to keep tidal volume in a safe range), peak inspiratory flow rates are commonly at lower during PCV than during *flow-controlled* VCV. To circumvent this problem, it is tempting to use higher levels of P_{SET} , guaranteeing a high peak flow at the beginning of a breath, but keeping inspiratory time short enough so that a high tidal volume is not delivered.³⁹ This maneuver, however, must be used under close monitoring, or combined with some inspiratory pause so as to avoid double triggering and “breath stacking.”

The algorithm configured for assisted PCV also may carry some advantages over PSV. First, in patients with unstable respiratory drive, the backup rate works as a safety and stabilizing mechanism, avoiding central apneas and sleep fragmentation, especially in patients with cardiac failure or in those receiving heavy sedation.¹⁵⁰ Second, although PSV is popular during noninvasive ventilation, serious concerns have been raised about potential dyssynchrony at end inspiration. During PSV, transition from inspiration to expiration is triggered by flow (i.e., when inspiratory flow decays below a certain threshold). Hence the presence of mask leaks or severe airway obstruction can render this mechanism inefficient—both conditions cause a relatively high inspiratory flow at end inspiration. Leaks simply mislead the inspiratory flow regulator of the ventilator, which cannot detect that inspiratory flow is not being delivered to patient. Conversely, severe obstruction changes the shape of inspiratory flow (see Fig. 9-7), making it less decelerating and pushing end-inspiratory flows closer to peak flow. With both conditions, the default expiratory triggering threshold for PSV (usually 25% of peak inspiratory flow) may never be reached, causing excessive prolongation of inspiratory time and patient overinflation.^{39,151–153} Under such conditions, pressure-controlled, *time-cycled* ventilation could be convenient: The intensivist needs only to adjust inspiratory time to match the patient’s spontaneous inspiratory time, usually 0.6 to 1.2 seconds.^{39,151,152} By changing from PSV to PCV, the intensivist necessarily decreases the freedom of ventilation pattern (restraining, for instance, the possibility of a spontaneous sigh and potentially increasing discomfort) but also avoids dangerous prolongation of inspiratory time. This is a difficult balance that has to be judged at the bedside.

During assisted PCV, especially under high flow demands, a fast “attack rate” to peak flow (at the initial phase) of the breath is desirable and decreases the work of breathing.³⁷ Thus, when changing from totally controlled (time-triggered) PCV to assisted PCV, it is advisable to increase the speed of pressurization (i.e., to decrease the *rise time* or to increase the inspiratory *pressure slope*).

Activation of Exhalation Valve during Pressure-Controlled Ventilation: Airway Pressure Release Ventilation

Opening of the Exhalation Valve

In theory, any pressure-controlled breath should limit airway pressures to P_{SET} independent of patient demand, dyssynchrony, premature expiratory efforts, or cough. In practice, however, hardware and software limitations of most ventilators preclude such ideal configuration. Usually, ventilators can overcome an increase in inspiratory-flow demands efficiently during PCV (caused by leaks or strong inspiratory efforts) by boosting flow through the demand valve. Airway pressures are raised quickly back to P_{SET} . Additionally, during the period in which the demand valve is still open, any sudden increment in alveolar pressure (cough) can be counterbalanced promptly by a sudden decrease in demand flow, attenuating the potential raise in airway pressures. Once the demand valve is already closed, however—because alveolar pressures have equalized airway pressures—the ventilator is no longer able to control airway pressures. Because most ventilators do not actively control the exhalation valve during inspiration (they just close it tightly until the end of inspiratory time), any unexpected increment in airway pressures can only be counterbalanced by total closure of the demand valve, but this is already at its minimum position. The end result is a steep increment in airway pressures, as in *flow-controlled* breaths.

Physicians have to keep this limitation of PCV in mind because it may be responsible for undesirable increments in airway pressures above P_{SET} in most ventilators. This situation has been observed during continuous tracheal gas insufflation,¹⁵⁴ during cough, and during strong efforts in assisted PCV breaths. In the latter circumstance, marked elevations of airway pressures can be found even in the absence of active expiratory efforts. This is so because the patient effort can increase effective driving pressures in early inspiration, increasing tidal volume and, consequently, the elastic recoil pressure at end inspiration (when diaphragmatic contraction already has ceased). An inactive exhalation valve could not release this extra tidal volume before the beginning of next exhalation, thus increasing airway pressures.

To overcome such limitations, some modern ventilators have incorporated improved hardware in their exhalation system that allows the simultaneous control of both valves (the inspiratory demand-flow valve and the exhalation valve) during the inspiratory phase of PCV. The end result is a smoother control of airway pressures during cough or during continuous tracheal gas insufflation.¹⁵⁵ Whenever available, such active control of the exhalation valve enhances ventilator operation.

The opening of the exhalation valve and its active control during inspiration are the technological basis for incorporation of APRV in microprocessed ventilators.

Airway Pressure Release Ventilation and Its Variants

APRV is a form of partial ventilatory support intended originally to offload a portion of the work required to ventilate during a primary crisis of oxygenation.^{143,144} In its original conception, APRV elevated mean airway pressure by maintaining a moderately high level of CPAP, delivered through a specially designed high-flow CPAP system linked to a release valve operated by a time controller. The system was designed to work independently of any commercial ventilator. Spontaneous breaths were planned to occur around this pressure baseline. Periodically, the airway was depressurized rapidly during one of the patient's deflation cycles, exhausting waste gas from the expiratory reserve before replacing it with fresh gas as CPAP rebuilt to the baseline level. Total time for both phases of the release cycle generally was brief, ideally occupying only a single breath of the patient's rhythm. The release pattern was repeated at a clinician-selected frequency. As predicated by the mathematics of pressure-preset ventilation,³³ ventilator support (achieved by the machine) was necessarily a function of the number of release cycles, the magnitude of the pressure drop during release, the duration of the release period, and the impedance to inflation and deflation. Synchronization between the patient's own exhalation and airway depressurization also affects the effectiveness of ventilation. Ineffectual inspiratory efforts during the deflation phase could impair the released volume. Despite such concerns, the original system was not designed to be synchronous.

Subsequently, different names were used for similar systems and arrangements, *biphasic positive airway pressure*, *intermittent mandatory pressure-release ventilation*, or simply PCV+ (this latter mostly used in the United States). Despite the similarity, the term *bilevel airway pressure*, commercially known as *biphasic continuous positive airway pressure*, represents something different, being simply a combination of pressure-supported ventilation and CPAP; this product is designed for noninvasive use. The basic difference among these systems is the duration of each phase, although all the systems can be approximated to periodic alternation between two CPAP levels according to a time controller. A longer period at the lower CPAP level, enough to allow one or more spontaneous breaths, is characteristic of biphasic positive airway pressure ventilation. This definition, however, is not consistent in the literature, with considerable ambiguity in the use of terms.²¹ Commercial ventilator branding further add to confusion. The specific transition from low to high CPAP levels differs among these variants, whether it is synchronized or not.

Adjustment of all these features is now possible in modern ventilators, and this does not represent any important conceptual modification. Accordingly, we believe that the confusing proliferation of names should be simplified under a single acronym: APRV.

In modern ventilators, provided that the ventilator works with an *active* (open) exhalation valve during inspiration, APRV can be perfectly imitated by PCV breaths with a single difference. During APRV, if the expiratory time is set long enough to allow the patient to complete one or more spontaneous breaths during the low-pressure phase, the spontaneous breath will be assisted by an efficient CPAP system (equivalent to PSV set at zero driving pressure). Conversely, during the exhalation phase of PCV, the spontaneous effort cannot be well assisted in most ventilators, because the bias-flow systems responsible for PEEP maintenance are not designed to support strong efforts. The end result may be some drop in PEEP unless sensitivity is adjusted to allow triggered inspirations (this latter, however, will result in an *assisted* PCV breath, which will necessarily cause an increase in the I:E ratio).

During the inspiratory phase, the opening of the exhalation valve during PCV assures that both systems perform identically, allowing spontaneous breaths to occur around highly sustained airway pressures. When spontaneous breaths are abolished by paralysis, APRV is not

different from PCV.

It is important to remember that spontaneous cycles and ventilator cycles are completely uncoupled in the most frequent usage of APRV. This means that spontaneous breathing can occur at any phase of the ventilator cycle without any synchronization mechanism. Switching between the two CPAP levels obeys a timed-cycled mechanism, and patient inspiratory work is mechanically supported only when it coincides with the restoration of the high CPAP phase. In view of such inherent dyssynchrony, more sophisticated APRV algorithms have been introduced in recent years, although without a clear rationale or clinical benefit. These include triggered transitions between low and high CPAP levels and use of PSV during one or both phases, on top of the corresponding baseline pressures.

Regarding the synchronized transition, it is important to remember that whenever commercial ventilators allow the patient to trigger the CPAP transition, usually during a time-window that spans the last 25% of the preset low-pressure phase, the actual time intervals at high and low pressures (equivalent to I:E ratio) may vary according to the respiratory drive of the patient.¹⁵⁶ Furthermore, even when some synchronization with CPAP transitions is attempted, most spontaneous breaths during APRV are still nonsynchronous because of the long duration of both phases. Under such conditions, the real advantage of synchronized systems is questionable. *From a workload perspective, APRV is an inefficient way of assisting the patient, imposing a higher workload and greater discomfort as compared with equivalent levels of PSV or assisted PCV.*^{65,139,157,158} The benefit, if any, of providing some synchronization has to be balanced against the loss of control over T_I/T_{TOT} . Synchronization would only make sense if applied to most spontaneous efforts, meaning that APRV should be set at higher frequencies and shorter inspiratory times, as with assisted PCV. Another potential disadvantage of synchronization is the creation of high driving pressures during this upward transition, caused by the coupling of muscle effort with the positive-pressure wave from the ventilator. The high inspiratory volume generated, however, will not be evident when looking at the next exhalation, because the release will occur against a higher PEEP level. To compute this hidden, large, inspired volume, one should carefully look at the next downward transition, a few breaths ahead, when most of the insufflated volume will be finally released.

Whenever APRV is applied in its common configuration, with long time intervals set for both phases, the intermittent mechanical assistance (applied only when the spontaneous effort coincides with transition from low to high pressures) shares some similarities with intermittent mandatory ventilation. As with synchronized intermittent mandatory ventilation, the presence of alternating levels of assistance probably impairs smooth accommodation of patient drive,¹⁵⁹ potentially causing some distress in patients when sedation is removed. Therefore, it is still an open question as to whether APRV will prove helpful during weaning or for patients with extreme weakness, very high workloads, or conditions in which release cycles are relatively inefficient in achieving ventilation (e.g., severe airflow obstruction).

For the sake of simplicity, whenever the patient gets too distressed during APRV and sedation is contraindicated, it is preferable to change to PSV or assisted PCV, assisting the patient in a more predictable way. The combination of PSV with APRV introduces unnecessary complexity, making little sense in terms of physiologic benefit or as an attempt to minimize driving pressures and ventilator-induced lung injury.

By extending the higher CPAP level further, APRV gets closer to inverse-ratio PCV. The shorter the time at low CPAP levels, the lower is the ventilator contribution to minute ventilation,¹⁵⁶ and the higher is the Pa_{CO_2} . On the other hand, Pa_{O_2} tends to increase because intrinsic PEEP increases. The full consequences of the duty-cycle settings on oxygenation are discussed in the section [Intrinsic Positive End-Expiratory Pressure versus Extrinsic Positive End-Expiratory Pressure](#).

Comparison of APRV associated *with* spontaneous breaths versus APRV *without* spontaneous efforts (identical to conventional PCV), when both modes are set at *equivalent end-expiratory* and *end-inspiratory* airway pressures, showed that the former achieved better oxygenation,¹⁴⁷ higher cardiac output,^{147,160} higher renal blood flow,¹⁶¹ higher lung aeration, and higher functional residual capacity.¹⁴⁵ Whereas the hemodynamic benefits seem to be related to lower intrathoracic pressures generated during spontaneous efforts,¹⁴⁶ the benefits in lung function seem to be mostly related to higher transpulmonary pressures.^{145,162} When equivalent transpulmonary pressures are accomplished by controlled PCV without any patient effort (with P_{SET} adjusted at higher levels, matching the transpulmonary pressures observed during APRV), the oxygenation benefits are no longer observed.¹⁶²

It is also important to stress that equivalent physiologic benefits can be achieved during simple CPAP application (matched to the high-pressure phase of APRV), in fact, with better outcomes in terms of oxygenation.¹⁶³ These findings demonstrate that *most of the benefits attributed to APRV relate to the replacement of mechanically applied driving pressures by patient-generated driving pressures*.

When considering the overall implications on hemodynamics, lung injury, and muscle function, APRV still poses a well-defined set of potential problems. Central vascular congestion and exacerbated pulmonary edema are natural consequences of lower intrathoracic pressures and increased cardiac workload. Interstitial pressures are expected to be lower in this setting, potentially increasing transcapillary pressures and interstitial edema.²⁶ Thus, in patients with cardiac failure, APRV should be used with caution. Conversely, to unload inspiratory muscles significantly, the high-pressure CPAP in APRV first must be raised to a considerable degree, obscuring the major advantage of APRV, which is the use of lower airway pressures. Moreover, inspiratory muscles will be disadvantaged by the resulting hyperinflation, promoting a sense of dyspnea that APRV is designed to relieve.

Finally, the intrinsic design of APRV raises important concerns about VILI. Because relaxation of inspiratory muscles is common during end-exhalation, even during spontaneous effort, we should not expect any significant advantage of APRV in terms cyclic lung collapse and its related damage.^{164–167} Also, because inspiratory transpulmonary pressures are unpredictable, probably higher than during passive PCV breaths (supposing a situation where P_{SET} matches the high CPAP level of APRV) we should not expect much improvements in terms of overdistension.^{18,43,168–170} Conceptually, the same principles of protective ventilation should be respected during APRV: the need to reduce effective driving pressures, whatever their source of generation, and the need to avoid cyclic collapse and tidal recruitment. The practical implications are that the releasing pressures of APRV should be titrated carefully (as during any PEEP titration), and, similarly, driving pressures should be estimated carefully and minimized (as during any attempt to reduce plateau pressures during controlled ventilation). Unfortunately, the general use of APRV rarely follows such protective principles.²¹

In conclusion, the important message provided by studies of APRV is that the preservation of spontaneous efforts is an important aspect of mechanical ventilation that should be employed prudently. Often, the simple use of assisted PCV, instead of fully controlled PCV, is the required change to promote profound benefits in hemodynamics, lung recruitment and oxygenation.

Combined Modes

In recent years, attempts have been made to combine the desirable features of both *flow*-controlled, volume-cycled and *pressure*-controlled, time-cycled ventilation. These modes include *pressure-regulated volume-control*, *volume-assured pressure-support ventilation*,²⁴ and *autoflow* among others; they guarantee any tidal volume compatible with the physician-set upper pressure limit. Some modes accomplish a dual target over a few breaths, whereas others, like volume-assured pressure-support ventilation, guarantee a dual target within the same breath. Despite limited research, the original rationale for these modes was the desire to combine the benefits of *pressure* and *flow* controllers while avoiding their problems.

A major advantage of such modes is their use in patients in whom some level of spontaneous ventilation is desirable, but in whom some control over transpulmonary pressures is necessary. Provided that the patient is not extremely agitated, a volume-cycled target promotes a decrease in inspiratory airway pressures in proportion to the magnitude of patient effort, still allowing the patient to choose inspiratory flow pattern. Such feedback limits transpulmonary pressures and, theoretically, would be a convenient strategy to avoid ventilator-induced lung injury. Also, such feedback usually provides a smooth transition from *ventilator-imposed* driving pressures to *patient-generated* driving pressures. The drawback of such strategy is the increase in the work of breathing, especially when the volume target is set below 6 mL/kg.^{125,127}

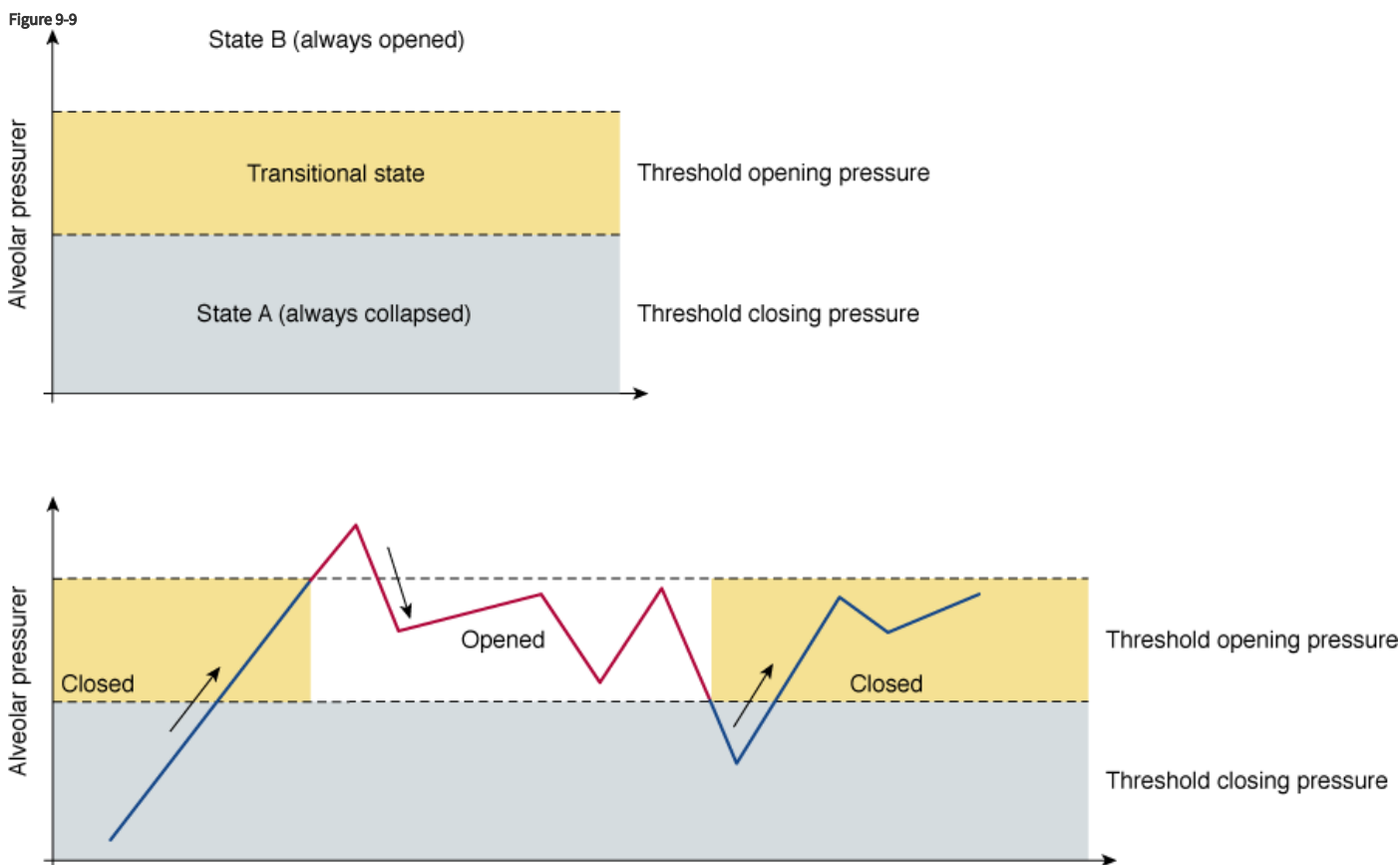
Inverse-Ratio Ventilation

After the tremendous enthusiasm that followed Reynolds' original publication in 1971,⁸ especially over the subsequent two decades,^{9–13,171–175} IRV has today become an almost extinct mode, with the exception of its incorporation into APRV (see the section [Activation of Exhalation Valve during Pressure-Controlled Ventilation: Airway Pressure Release Ventilation](#)). Most studies on IRV used duty cycles (T_I/T_{TOT}) between 0.5 (the lower limit consistent with its definition) and 0.8. The original concept behind IRV was the attempt to increase $\bar{P}_{aw}$ and $\bar{P}_A$ without violating the desired upper limits for airway and alveolar pressures.

In initial studies, IRV was implemented either with *pressure-controlled*, time-cycled forms of ventilation or with *flow-controlled*, volume-cycled ventilation. Over time, the pressure-controlled application gained wider acceptance because of its better performance in terms of gas exchange^{28,176} and the lower risks of over inflation when paralysis was not total. Specific details and advantages of those earlier implementations can be found in Chapter 10 of the second edition of this book.

The enthusiasm associated with IRV was driven by the original report of three main physiologic advantages: (a) oxygenation improvement despite the use of low external PEEP,^{9-12,171} (b) apparently lower peak inspiratory pressures for equivalent tidal-volumes,¹⁷⁷⁻¹⁷⁹ and (c) very low levels of dead space and efficient CO₂ removal.^{72,76,176,177,179-183} Over time, it became clear that these advantages had a high price, and that we have better alternatives while maintaining conventional I:E ratios (<1:1).

The oxygenation benefits of IRV were shown to be the result of two basic mechanisms: (a) higher $\bar{P}_{aw}$ and $\bar{P}_A$ (when compared to conventional I:E ratio at same levels of external PEEP and plateau inspiratory pressures), and (b) higher intrinsic PEEP, promoted by the shorter exhalation time. Other mechanisms, like better gas distribution, were initially proposed and later found to be irrelevant. The relative efficacy of these two basic mechanisms was shown to be related to lung history and the degree of lung *hysteresis*. Some of the intricate relationships between the two mechanisms are shown in [Figures 9-9 and 9-10](#), which can be used as a guide to understand most studies on IRV (even those suggesting an apparent controversy).

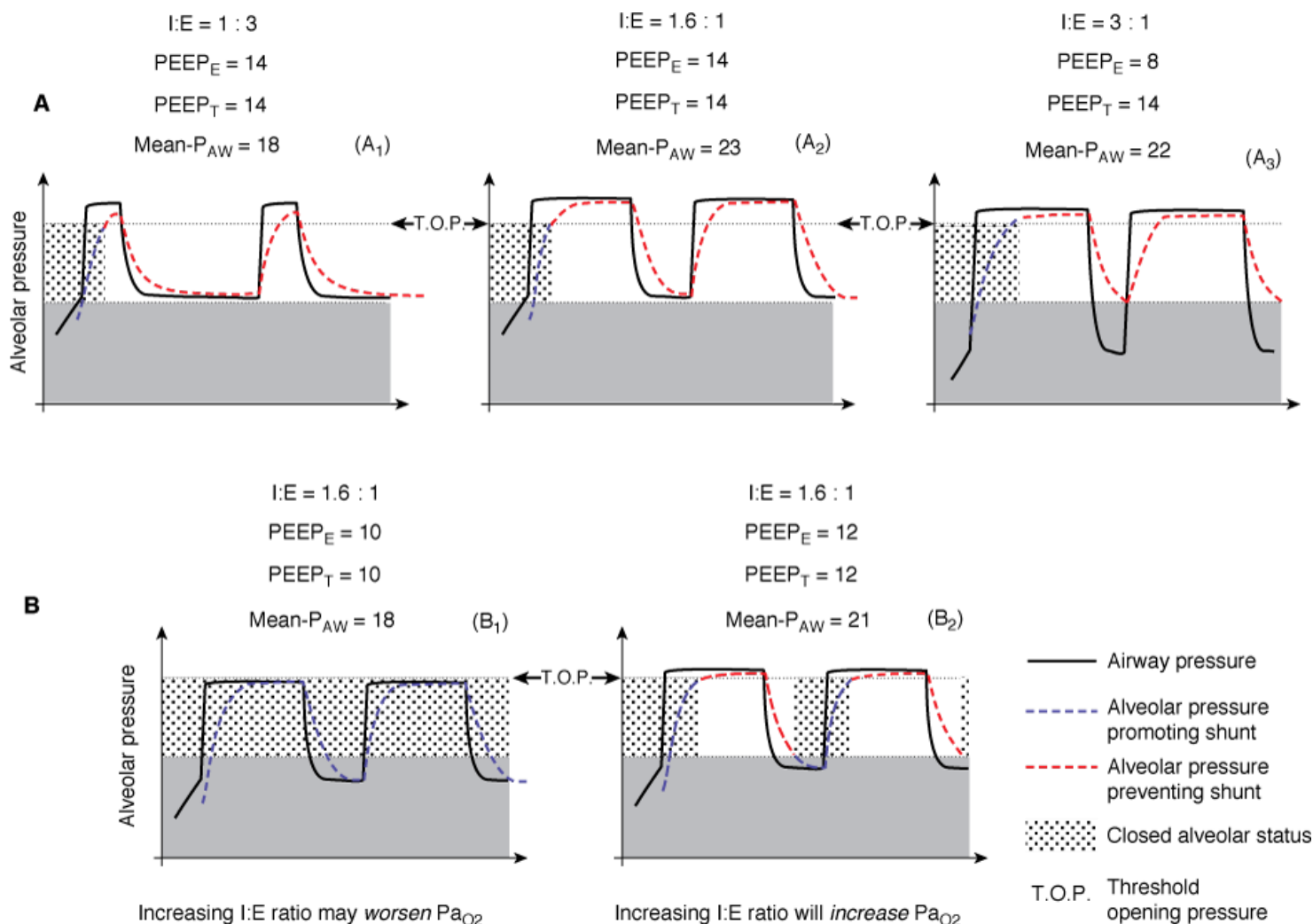


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Representation of lung hysteresis according to the pressures applied on the system (for simplicity, we considered alveolar pressures as roughly representing transpulmonary pressures). The system changes its status according to the direction (in time) of the transition. The transitional state keeps some “memory” of the previous state, behaving like in *state A* when coming from lower pressures, or behaving like in *state B* when coming from higher pressures. Once inside the transitional state, the situation can be stable for a long period, provided that pressures do not trespass the boundaries, that is, the threshold opening/closing pressures. The *red color* illustrates an alveolar condition producing low pulmonary shunt (perfusion of alveoli in an open condition), whereas the *blue line* illustrates a condition associated with high pulmonary shunt (perfusion of closed alveoli).

Figure 9-10



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A. Low-shunt situations. Three ventilator settings producing the same effect on Pa_{O2}. In each, alveolar pressures exceeded the threshold opening pressure after the first breath *and* were kept above the threshold closing pressures (defined by the gray zone) for the rest of cycles. Provided that a major part of lung units is adequately represented by the boundaries of threshold pressures indicated, the three settings will result in very low and equivalent shunt levels. Under such conditions, increments in mean airway pressure (produced by increased I:E ratios, as in the examples A₂ and A₃) will only put hemodynamics at risk. **B.** High-shunt situations. Because *end-expiratory* alveolar pressures are now trespassing on the closing pressures, the situation changes dramatically. The effect of increased mean airway pressure differs according to *end-inspiratory* pressures developed. In B₁, because P_{SET} is below the threshold-opening pressures, prolongation of inspiratory time can only impair hemodynamics, but the lung stays closed and producing high shunt levels. Conversely, if P_{SET} and alveolar pressures trespass threshold opening pressures during inspiration (B₂), Pa_{O2} becomes dependent on mean airway pressure. The longer the inspiratory phase, the longer the open status and the higher the Pa_{O2}. This situation is commonly observed when lung hysteresis is low or, alternatively, when driving pressures are relatively high, producing tidal recruitment/derecruitment. Alveolar pressures are represented in blue or red dashed lines, whenever they are promoting or preventing pulmonary shunt, respectively.

The major lesson to be learned from the studies on IRV is that lung hysteresis plays a greater role in the oxygenation response, irrespective of the I:E ratio, than was initially thought. Irrespective of its basic mechanism—which is linked to a complex sequence of lung opening and closing at the microscopic level—hysteresis can be generally understood as a lung *parenchymal memory* to pressure challenges. As schematized in Figures 9-9 and 9-10, one can regard hysteresis as representing an asymmetry of lung status or configuration according to the direction (in time) of pressure variations across the lung. Hysteresis means that there is a transitional state (representing a corresponding window of pressures) in the parenchyma that keeps “memory” of the previous state; if many lung units were closed in the previous state, it is likely that the system will stay partially or totally closed during subsequent increments in pressure. It is only when pressures reach a certain limit (*opening pressures*), defined by the transitional window of pressures, that we can see a change in status. And the opposite is also true: a

previously opened lung state will favor an open lung state during subsequent application of airway pressures, even if airway pressures are brought down to below the opening pressures.

The range of such a transitional window of pressures depends on the distribution of opening and closing pressures of lung units. The higher the mean threshold opening pressure and the lower the mean threshold closing pressure, the larger is this transitional range, and the more the entire system depends on lung history. If we assume, for the sake of simplification, that the blood flow through a lung unit does not change much from inspiration to exhalation, we can grasp the consequences of a large hysteresis on pulmonary shunt and, consequently, on oxygenation.

If hysteresis is large, with a wide transitional window, it is likely that inspiratory plateau pressures will remain below *opening pressures* in a large portion of the lung. In this scenario, a single recruitment maneuver might have a profound impact on those units and, consequently, on oxygenation. Symmetrically, the subsequent use of a proper PEEP, by keeping (or not) alveolar pressures above the threshold closing pressures, would also have a profound impact on oxygenation, preserving (or not) the new status achieved by the recruitment maneuver. In clinical practice, the range of this transitional window of pressures (representing hysteresis) exceeded 30 to 40 cm H₂O in patients with severe ARDS or trauma.^{15,56} Such a large hysteresis explains the apparent discrepancy in oxygenation responses across many studies on IRV. A careful match of previous pressure challenges, as well as a careful match of subsequent total PEEP application, is a prerequisite for any fair comparison. The same principles must be applied today in our understanding about the effects of recruiting maneuvers and external PEEP.^{15,184,185}

Conversely, if hysteresis is small, and by using the same rationale, it is also possible to explain why changes in $\bar{P}_{aw}$ (as achieved through prolongation of inspiratory time) are linearly related to oxygenation in particular settings. In fact, a small hysteresis favors the occurrence of tidal recruitment, because it is easier for alveolar pressures to exceed *opening pressures* during inspiration and, similarly, to reach *closing pressures* during exhalation (in the absence of hysteresis, *opening pressures* would equal *closing pressures*). Under this circumstance, any prolongation of inspiration would prolong blood flow through open lung units, decreasing shunt fraction. Using ultrafast intraarterial partial pressure of oxygen (Po₂) sensors, Baumgardner et al¹⁸⁶ showed that arterial Po₂ can oscillate by more than 400 mm Hg along the respiratory cycle, demonstrating the importance of the intricate relationships between *opening and closing pressures* and its direct influence on oxygenation.

Nonetheless, as explained in [Figure 9-10](#), it is often possible to achieve the same oxygenation with conventional I:E ratios, while generating a much lower $\bar{P}_{aw}$ and, consequently, milder hemodynamic consequences.^{72,73,176,181,182,187} The latter observation means that IRV represents a waste of $\bar{P}_{aw}$ application: it is always possible to achieve the same oxygenation benefits at a lower hemodynamic cost while using a shorter inspiratory time. More than a waste of airway pressures, some experimental reports have shown greater lung injury and edema linked with longer distending stress during inspiration.^{101,187}

The lower peak inspiratory pressures achieved during IRV¹⁷⁷⁻¹⁷⁹ were never proved to be consistent. Many studies show a decrease in peak airway pressures,^{72,76,176,181-183,187} but not in peak alveolar pressures. In the few studies in which a true reduction in plateau inspiratory pressures was achieved by IRV (associated with lower levels of total PEEP), the oxygenation benefits could no longer be demonstrated.¹⁷⁷⁻¹⁸⁰

Finally, the better CO₂ removal by IRV^{72,76,176,177,179-183} was shown to be of minor importance and related to the decelerating flow pattern, which increases the *mean distribution time* of the inspired gas.^{41,42} As explained in the section [The Controversy on Optimal Distribution of Ventilation](#), however, most of this effect is already observed when T_I slightly exceeds the point of zero flow (see [Fig. 9-6](#)), which means that prolonging inspiration beyond this point is of questionable benefit.

Intrinsic Positive End-Expiratory Pressure versus Extrinsic Positive End-Expiratory Pressure

The research on IRV has fostered great controversy on the role of intrinsic PEEP, as opposed to extrinsic PEEP, in patients with alveolar instability. According to some initial proponents of IRV, intrinsic PEEP (intentionally promoted by IRV) should be considered as a “selective” form of PEEP that preferentially ventilates parts of the lung with short time constants, placing at rest parts with prolonged time constants.¹⁸⁸ In a theoretical exercise, if respiratory rate approaches infinity, alveolar pressures would remain practically constant in the *slow* units, equilibrating at an intermediate value between PEEP and P_{SET}. Because healthier units tend to have slower time constants and to be located

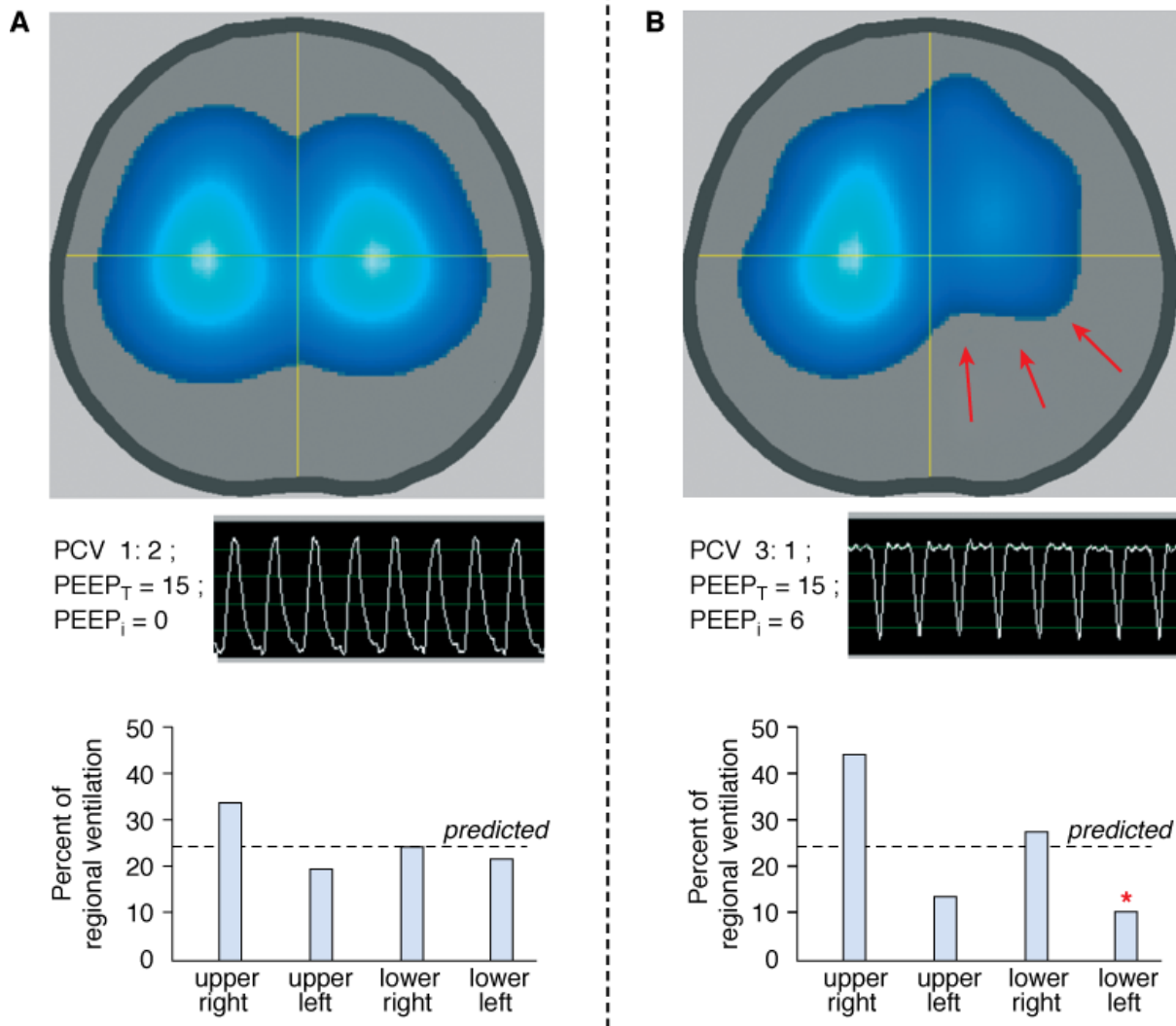
in nondependent regions in patients with ARDS, intrinsic PEEP is expected to produce less cyclic overdistension of the nondependent parenchyma, increasing ventilation in dependent and more perfused lung regions. Consequently, such selective improvement of ventilation at low $\dot{V}/\dot{Q}$ areas would eliminate CO_2 more efficiently.¹⁸⁹ Another way of formulating such rationale has been to propose a strategy in which “the absolute time of the expiratory phase is so short that the stiffest parts of the lung have no time for collapse.”¹⁸⁸ Thus, external PEEP could be reduced, with desirable benefits in terms of hemodynamics.

Recent evidence does not support this view but demonstrates the opposite.^{54,76,190–194} Because different time constants in individual lung regions are crucial for a “selective effect” of intrinsic PEEP, intrinsic PEEP may allow the “selective collapse” of regions with short time constants, especially the edematous regions. When comparing intrinsic PEEP with matching levels of external PEEP in patients with ARDS, Zavala et al⁷⁶ demonstrated that shunt levels increased by 6% with intrinsic PEEP. There was some increase in the efficiency of CO_2 removal, but this could be ascribed to better inspiratory flow distribution during IRV rather than to intrinsic PEEP. Animal experiments in dogs¹⁹¹ reproduced the same results. Other clinical studies came to similar conclusions,¹⁹³ including studies in patients with obstructive disease,¹⁹⁴ suggesting that partial or complete substitution of intrinsic PEEP by external PEEP improves gas exchange without any impairment in CO_2 removal.

The original perspective of considering absolute time for collapse¹⁸⁸—instead of end-expiratory pressures—also carries some problems. Computed tomographic studies demonstrate that the time window to prevent end-expiratory lung collapse without external PEEP is extremely short (<0.6 second) and might not be feasible during mechanical ventilation of many patients with ARDS.^{54,192} To avoid unnecessary confusion, it is also important to prove that time per se, independent of local alveolar pressure, is an important variable affecting lung collapse. To our knowledge, this has never been demonstrated.

After reviewing available evidence, we conclude that there is no rationale supporting the preferential use of intrinsic PEEP. In terms of oxygenation, it seems inferior to external PEEP, slightly increasing the risk of collapse in unstable units.¹⁹⁵ In lung models with different time constants, it is clear that fast (more disease) units are at a disadvantage during intrinsic PEEP generation.¹⁹⁰ Figure 9-11 provides further insight: regional ventilation becomes more heterogeneous with intrinsic PEEP, with evident collapse in the left lower lung, and ventilation is increased in nondependent regions. A computed tomographic study revealed more hyperinflation with intrinsic PEEP than with equivalent external PEEP.⁷⁴

Figure 9-11



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Regional ventilation distribution measured by electrical impedance tomography (EIT) in an animal model (pig) of acute lung injury. There was preferential injury of the left lung. The functional images display the amplitude of cyclic changes in aeration for each pixel during tidal ventilation. Bright colors represent regions with higher swings in aeration and hence higher regional ventilation. Perfusion perturbations were subtracted by temporal filtering. Although total PEEP in both conditions was the same (15 cm H₂O; checked by end-expiratory occlusion), the left image was obtained during ventilation with external PEEP equaling 15 cm H₂O and PCV with I:E ratio equaling 1:2, whereas the right image was obtained during 9 cm H₂O of external PEEP added to 6 cm H₂O of intrinsic PEEP (PCV with I:E ratio = 3:1). This latter pattern resulted in more heterogeneous ventilation, with preferential ventilation of the right lung and marked hypoventilation of the left lower lung (*red arrows* and *asterisk*). The black midpanel represents the temporal changes in total impedance of the thoracic slice monitored by EIT, which roughly illustrates the cyclic lung volume changes associated to each condition. The graph with bars at the bottom shows that external PEEP produced more even pattern of regional ventilation.

In terms of CO₂ removal, it is impossible to draw any conclusion because the concomitant use of IRV and decelerating flow in most studies precludes any estimation of the isolated impact of intrinsic PEEP. In clinical practice, one also has to consider that a combination of IRV plus intrinsic PEEP usually results in a higher $\bar{P}_{aw}$ than conventional I:E ratios plus external PEEP and thus greater hemodynamic compromise. Another major drawback of the intrinsic PEEP approach is its complexity. Much closer monitoring and more frequent adjustments of ventilator settings are required, even with a computerized protocol.¹⁹⁶ Consequently, external PEEP in association with conventional I:E ratio is preferable.

If CO₂ removal is a clinical problem, an increased frequency—even if associated with low tidal volumes—may solve the problem, especially when using high PEEP levels. By increasing functional residual capacity, external PEEP usually causes a decrease in airway resistance and

lung compliance, both factors contributing to a very short time constant. Under such conditions, respiratory rates up to 40 to 50 cycles per minute could increase minute ventilation considerably (see [Fig. 9-3](#)), improving CO₂ removal.

The Future of Pressure-Controlled Ventilation

During controlled ventilation, there is little doubt that **PCV** is advantageous in terms of lung protection, helping physicians better define the targets for lung protection. This is especially true during recruitment maneuvers or in any situation where lung tissue is close to its limits of rupture. In this context, we believe that the growing acceptance of **PCV** over VCV represents a consistent trend. In the presence of spontaneous efforts, however, the future of assisted **PCV** is still uncertain and we need more studies to solve important physiologic questions. For instance, during protective, low tidal-volume ventilation, the assisted-PCV option commonly result in higher transpulmonary pressures and higher work of breathing than occurs with optimally adjusted assisted VCV.²⁵ This is not a limitation of the mode per se, but a problem in the way we employ it. This is also true when we have to choose between assisted **PCV** and PSV to improve synchrony. Research that addresses deeper questions about priorities in such situations will help us to better define the future of assisted-PCV.

Summary and Conclusion

Contrasted with the growing acceptance of **PCV**, use of IRV is almost extinct. Safer, more predictable, and more efficient ventilator solutions can be implemented at the bedside. Physicians were initially enthusiastic about IRV because internal PEEP and peak pressures were hidden. Later, physicians realized that total PEEP and peak alveolar pressures were not that different, yet IRV necessarily imposed higher levels of $\bar{P}_{aw}$ and greater hemodynamic consequences for similar clinical benefits. Whereas concepts such as VILI and permissive hypercapnia have boosted the use of **PCV**, the better understanding of the dynamics of alveolar collapse have dampened enthusiasm for use of IRV.

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